

SHIPS AT WORK



MARY ELTING
ILLUSTRATED BY
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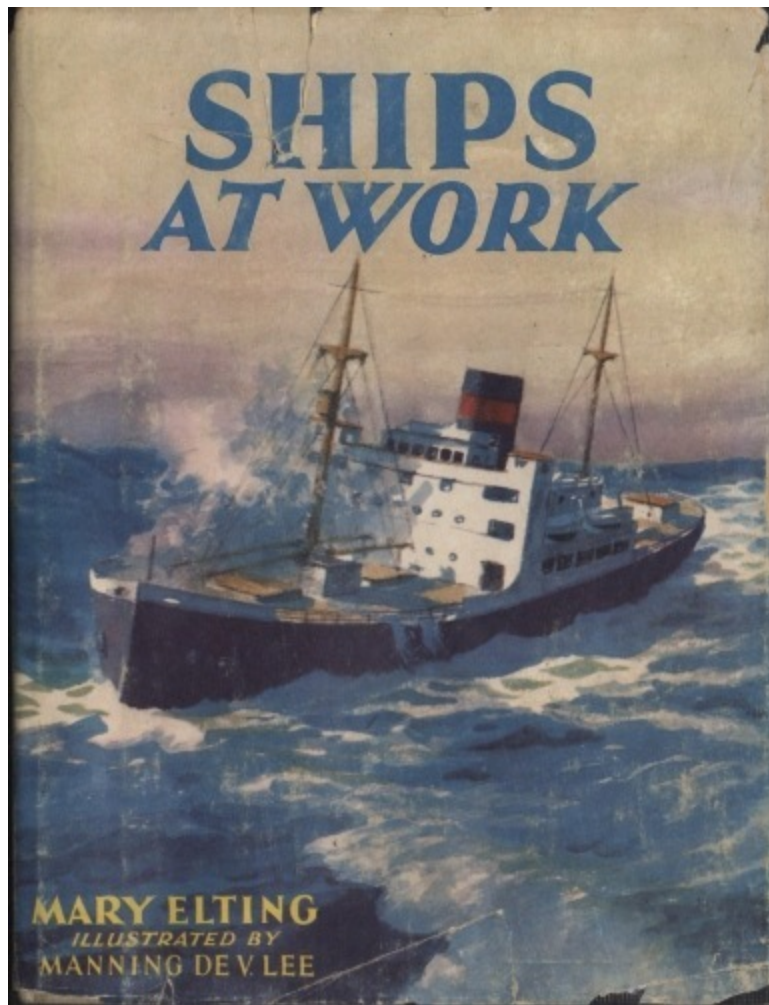
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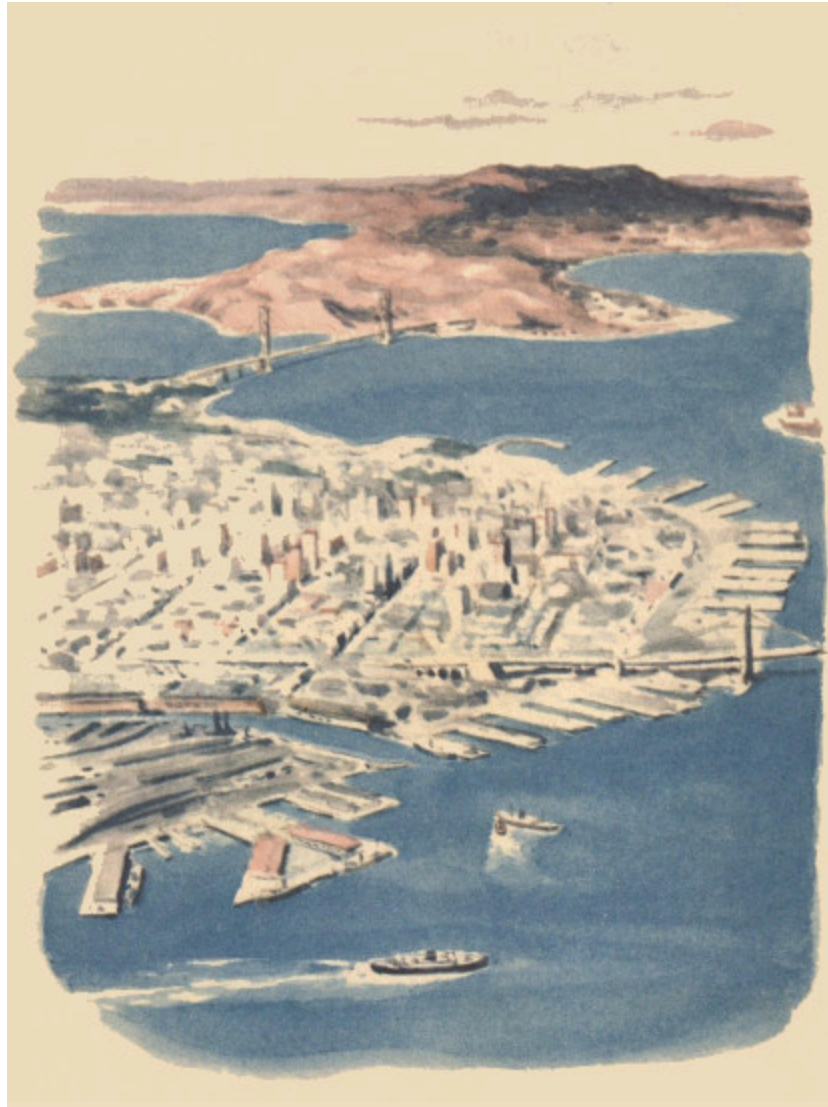




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SHIPS AT WORK





SHIPS AT WORK

By Mary Elting



ILLUSTRATED BY
MANNING DE V. LEE

GARDEN CITY BOOKS GARDEN CITY,
N.Y.



SHIPS AND MEN

A ship is a marvellous thing. It took ships—and the men who sail them—to circle the world and tie it all together into one round ball. Brave seamen from a thousand ports have faced storms and unknown dangers, first to make the world a bigger place for people to live in, then to bring all people close together.

No matter how dangerous the voyage nor what she carries, a ship is always “she” to a seagoing man. He never calls a freighter or a tanker or any large vessel a boat. Only shoreside people who have never been to sea make the mistake of calling a ship a boat. And shoreside people never know the excitement and fun—and the long, hard work—that the skillful men of the sea know every day of their lives.

STANDING WATCH

Jim is a sailor on a freighter carrying cargo across the Atlantic Ocean. Every morning at half-past three, someone comes into the forecastle. That's the seamen's name for their sleeping quarters. They pronounce it "foke-sull."

Jim mumbles a little. Then the light goes on. The sailor who has waked him wants to be sure he doesn't go back to sleep. With half-open eyes, Jim sees his clothes hanging from hooks. Back and forth they sway as the ship pitches and rolls. Jim is so used to sleeping in rough weather that he hadn't even noticed when a storm blew up in the night.



Now he's wide awake, and so are the other men in the forecastle. Jim swings his legs over the side of his bunk, in a hurry to get dressed in well-washed blue dungarees, a turtleneck sweater instead of a shirt, thick socks and a heavy woolen pea coat. That's a sailor's winter jacket with pockets that slant in sideways. He makes sure his sharp knife is dangling from a snap on his belt. No telling when it might come in handy. Then he sticks a knitted blue stocking cap on his head and reaches for his fleece-lined mittens.



Jim wants to be warm. He knows the wind will be sharp, even though his ship is headed for the warm Mediterranean Sea. It's wintertime and still cold out on the Atlantic Ocean.

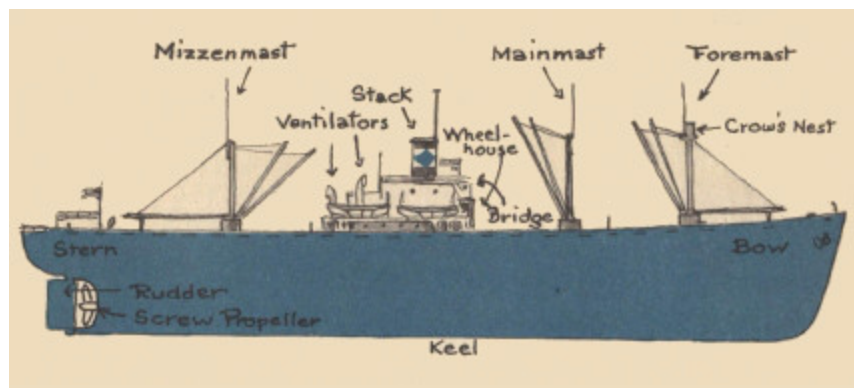
Jim and the three men who share his bunkroom are ready for work—almost ready. First they go down the passageway to the mess, which is their word for dining room. There they have coffee from a big steaming urn that is always kept full and hot. In another minute Jim steps out onto the leeward side of the deck—the side away from the wind. Although he's in a hurry, he waits there sheltered from the wind for a few minutes while his eyes get used to the dark. Jim is going to stand his watch. That means he will work for four hours.

Jim is an AB—an Able Bodied Seaman. An AB works out on deck instead of down inside the ship in the engine room or in the kitchen, which he calls the galley. All the men who work on a ship are seamen. Only deckhands are called sailors. And only those sailors who have passed examinations and have been at sea for a certain length of time are AB's. The other sailors are called ordinary seamen or ordinaries for short.

As soon as his eyes can see in the dark, Jim walks toward the bow which is the front of the ship. As the deck rises and falls and tilts under his feet, he manages from long practice, to keep his balance, but he also slides one hand along the rail on top of the bulwark, a kind of low wall that runs all around the deck.

In good weather he would go to the bow and stand there, watching for anything there might be in the ocean ahead. But tonight waves may splash over the bow. An unexpected wave can knock a man down or even wash him overboard. It will be safer high up in the crow's nest above the deck.

Besides he can see farther from up there. So Jim climbs to the little enclosed platform high on the foremast.

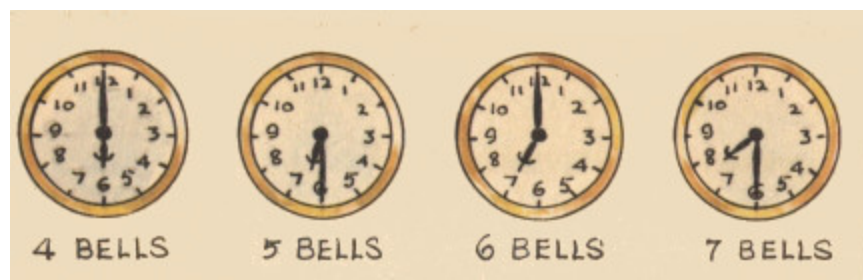


In a very bad storm Jim would not go outside. He would stand watch in the wheelhouse. This is a room with a big window high above the deck in the part of the ship called the house. The room gets its name because the wheel that steers the ship is in it.

Jim knows it is good manners always to be a little early when you go to take the place of another seaman whose watch is over. So he doesn't waste any time as he scrambles up the steel rungs in the ladder on the mast.

He pokes his head through the hole in the floor of the crow's nest. There he finds Juan, who is cold and glad enough to climb down and get into his warm bunk.

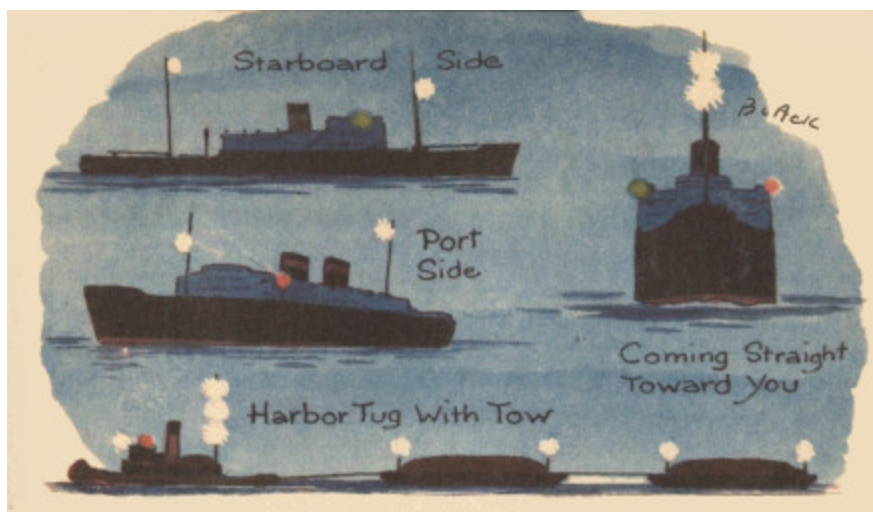
Juan has a telephone strapped on his head. He uses it to talk with the third mate, the officer in charge of the ship who works in the wheelhouse. When Juan sees Jim, he says into the telephone, "Crow's nest to wheelhouse—being properly relieved, sir." Now the mate, listening to the loudspeaker in the wheelhouse, knows that Jim is the lookout in the crow's nest.



Jim puts the telephone on his head and leans against the rail around the small platform that sways far to one side, then to the other. Soon he hears the ship's bell, a faint sound above the storm—"Ding-ding, ding-ding, ding-ding, ding-ding." Eight bells. It is exactly four o'clock. At four-thirty the bell rings again, just once. Two bells will be five o'clock, and so on until eight, when there will be eight bells again.



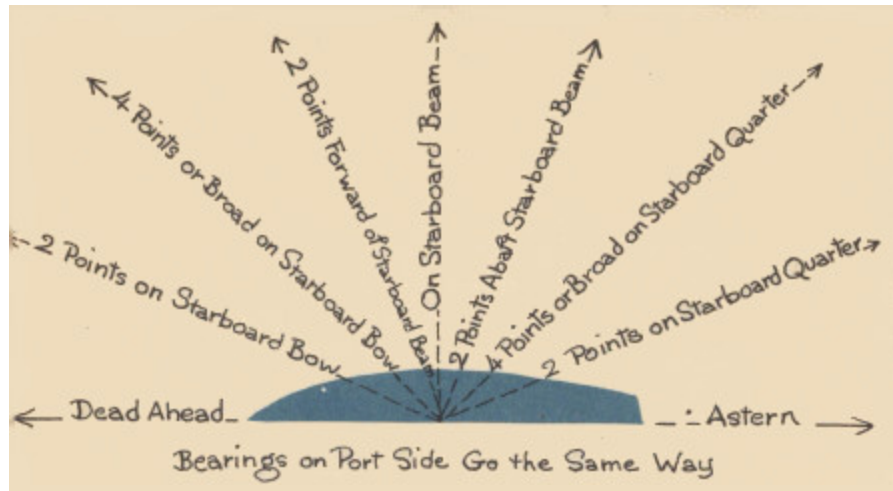
For a long time there is nothing for Jim to see but great gray waves rising and lifting the ship, and once in a while splashing over the decks way down below. Then far ahead and to the right Jim sees a tiny speck of light.



"Crow's nest to wheelhouse," he calls into the phone. "White light two points on the starboard bow." The mate knows from this where to look for the light. The diagram on page 16 shows the words Jim will use when he tells the mate to look in other directions.

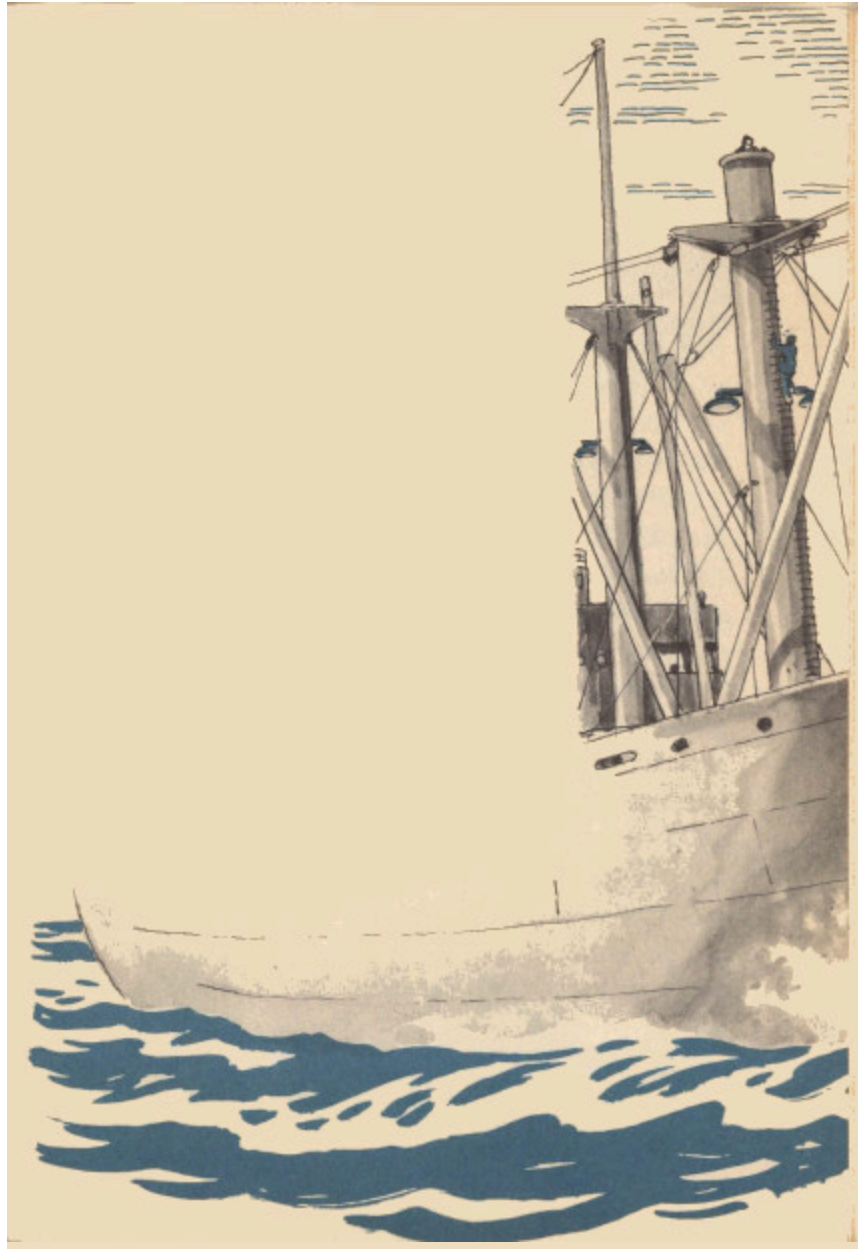
Jim thinks the white light probably comes from another ship. Soon he knows it does. He can see two white lights very close together and a green light a little below them. He and the mate know that a green light is always shown on the right or starboard side of a vessel that's moving. There is no

danger. Jim's ship and the other one are a long way apart and are not headed for each other. If Jim saw both a green light and a red light with two white lights above them, he would be alarmed. This would mean a ship coming straight at him.



Now and then spray from the waves blows all the way to the crow's nest, and Jim is glad of a protecting shield that comes up almost as high as his face. But he can feel the wind anyway, and he can hear it roar through the rigging. He almost has to shout into the phone so the mate can hear him.

The safety of the ship depends on Jim. Even in the darkness he can see a great deal from his high perch. He may notice the white foam of waves ahead behaving in a strange way. This could be the wreck of a half-sunken ship that would tear a hole in his own ship and send her to the bottom. If he dozed off, he might fail to sight some danger. So he must keep alert every minute. He's responsible for the lives of all his shipmates, and he takes his job seriously.



Jim watches the dark, heaving ocean for two hours. He's glad when his coffee time comes. That's ten minutes of rest he gets after standing watch for two hours. When another lookout comes to the crow's nest to take his place, he warms up in the mess and then goes to the wheelhouse. There he works for two hours steering the ship. He stands his watch at the wheel.

The wheelhouse is dark, so that the mate can see through the big windows anything that the lookout reports. The only light comes from instruments, such as the compass. Jim watches the compass to make sure he is steering in the right direction. The mate tells him what direction the

captain has ordered the ship to go. But the compass can't be their only guide.

When you guide yourself by a compass on a hike across a wide meadow, you can keep going in a straight line because nothing pushes you to one side or the other. But at sea the wind is always pushing against a ship, making it slip sideways. Currents in the water push, too. The current may be going one way and the wind in another. There are no trees or mountains on the ocean to help seamen know exactly where they are. So they can use the sun and stars as their guides.

Of course, the sun, stars and moon keep moving. But they travel in an orderly way. If a seaman knows the rules about their motion, he can look at them through special instruments and figure out where he is. He can navigate.



More than two hundred and fifty years ago, an American boy named Nathaniel Bowditch went to sea and discovered that sailors didn't have any good, accurate rules for steering by the stars. He decided to do something about the problem. Before long he had worked out a set of rules that were so good that every man in his crew could navigate—even the cook!



The mate on Jim's ship has instruments with which he looks at the sun and stars. And he still uses the book that Nathaniel Bowditch wrote so long ago.

Besides the wheel and the compass, there are other instruments in the wheelhouse. One is the engine room telegraph. The mate uses this when he wants the ship to go faster or slower, forward or backward. He moves the handle of the telegraph, and a bell jangles in the engine room. Another telegraph there, exactly like the one in the wheelhouse, shows the engineer at what speed the ship should go. To let the mate know he has received the order, the engineer sends the same signal back on the telegraph, and a bell in the wheelhouse jangles, too.

By eight o'clock, when it is daylight, Jim's watch is over. He goes below, as seamen say, and sits down with his messmates—all the others in the crew who aren't on watch—for a big breakfast of orange juice, bacon, eggs and flapjacks. Then he goes to sleep.

A little before noon he is up again. The storm was not a bad one. The sun is shining, and it is warm out on deck. Jim has all afternoon until four o'clock to himself. This is how he spends it: First he gets a bucket of cold water and puts it under a little faucet that brings up steam from the engine room. He runs steam into the water, and it's hot in a few seconds. Out on the afterdeck, sailors have rigged up a washboard.

Jim spreads his dirty clothes on the board and scrubs them with a brush and soap and his steam-heated water. Seamen do a lot of washing. They like to keep their clothes clean. Often they do their own mending, too.



While Jim's clothes dry on a regular clothesline on the afterdeck, he gets out his ditty bag which holds all kinds of odds and ends, including needles and thread and a sailor's palm. The palm is what a sailor uses instead of a thimble for pushing a big needle through heavy canvas. In the old days when ships had sails to be mended, these palms were very necessary, but nowadays most sailors only use them the way Jim does. He is making a sea bag to take the place of his old one that has worn out. The sea bag is his trunk. He carries it on his shoulder whenever he changes ships.



While Jim sews, he sings, and other seamen who are off watch sing too. One of them plays a banjo, and another has a harmonica. Some of the songs are the ones you hear any day on the radio, and others are songs that seamen themselves have made up.

These sailor songs are called chanteys—pronounced shantys. On old sailing vessels men sang them as they worked together, and the rhythm of their work set the rhythm of the music. Here is a chantey that helped them pull together on the rope that lifted a sail:

Way! Haul away! We'll haul away the bowline.
Way! Haul away! We'll haul away, Joe.

In those days, before there were engines to do work, men used a hand-turned machine called a capstan to raise the anchor or tighten heavy lines. They turned it round and round by pushing against long bars called capstan bars. As they pushed, they sang:

Yo, heave ho! Round the capstan go.
Heave, men, with a will. Tramp, and stamp it still!
The anchor must be weighed, the anchor must be weighed.
Yo-ho! Heave ho! Yo-ho! Heave ho!

Now, while the singing goes on, Jim takes his turn at having a haircut. For a barber's chair he uses a bitt. That's a round piece of steel that sticks up out of the deck at just the right height. It's used at times for holding big ropes that seamen call hawsers.

The barber is a man from the black gang. That means he works in the engine room. When he is off watch, he likes to make a little extra money cutting hair. So he puts a sheet around Jim and starts to work. Chiquita, the ship's cat, takes a playful swipe at a dangling corner of the sheet, and then goes off in search of a rat that may have come aboard in port.



The barber has pictures tattooed on his forearms, and Jim laughs as he watches them. On one arm is a picture of an old sailing ship. As the barber's muscles move, they make the ship look as if wind is blowing on the sails. On the other arm is a beautiful lady chasing butterflies. When the barber opens and closes the scissors, the lady looks as if she is dancing after the butterflies.

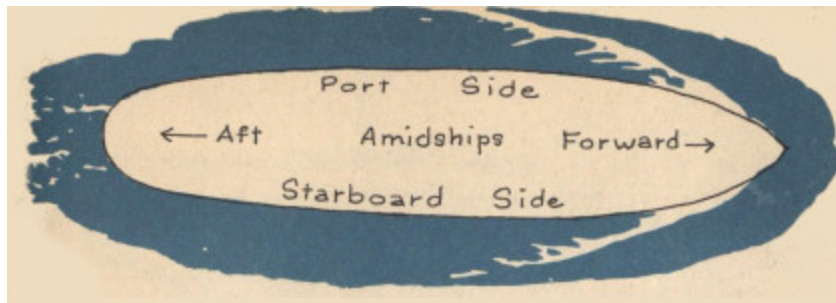
Just before four o'clock, Jim goes to mess again. Then he's on watch for four more hours to put in the rest of his eight hours of work in a twenty-four hour day. He stands lookout again for two hours and takes the wheel for two more. Now his day is done.

SEA LANGUAGE

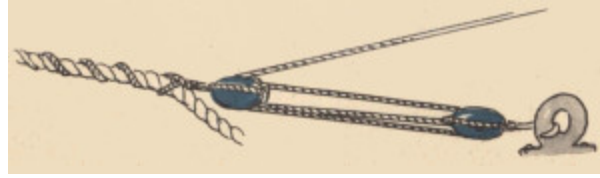
When Jim first went to sea, he found that seamen speak a language of their own. A floor is always a deck. A partition between rooms is a bulkhead. A ceiling is the overhead. Stairs are always a ladder. The opening onto a deck at the head of the steps is a companionway. Almost all ropes are called lines.

One day another seaman said to Jim: “The bosun wants you to break out the handy billy in the forepeak and take it aft to Chips. He’s abaft the mizzenmast.” This is what all those words mean:

The bosun is a man who acts as foreman, giving orders to deckhands. “Break out” means “take from its regular storage place.” The handy billy is a combination of small wheels called blocks with a line running around them. It is handy for moving heavy weights. The forepeak is a storeroom under the main deck at the bow where the bosun keeps tools and equipment. Chips is the ship’s carpenter. Aft means toward the stern of the ship, and abaft means “behind, in the direction of the stern.” The mizzenmast is the third mast, counting from bow to stern.



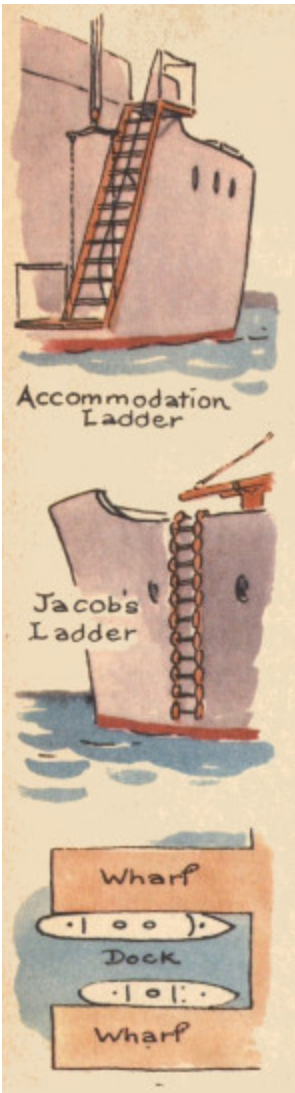
Jim also had to learn that anything toward the bow of the ship is forward. Anything toward the middle is amidships, and anything crosswise is athwart or thwartships. Anything on the windy side of a ship is to windward. (A good sailor never spits to windward.) Anything on the side away from the wind is to leeward—pronounced “loo-urd.” When Jim goes up on deck he goes topside; when he climbs a mast, he goes aloft.



Jim had to learn the commands that the mate gives him when he is at the wheel steering the ship. Helm is another word for the wheel, and helmsman is the man who steers. (On some ships, Jim would not steer at all. Steering is often the special job of AB's called quartermasters who don't do much of anything else.)

Suppose the mate says to Jim, "Mind your rudder." That means Jim must steer carefully or get ready for a new order. "Steady as you go" means keep on going just as you are.

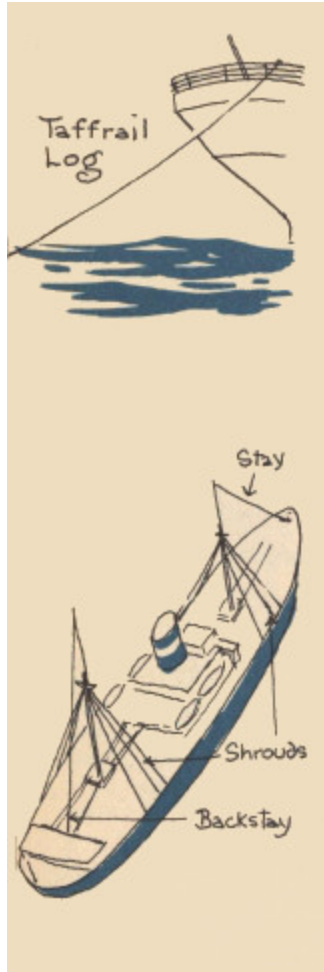
The wheelhouse is sometimes called the pilot house. The pilot is a man who specializes in guiding ships in and out of harbors. A small boat brings him out from shore. Usually he climbs aboard on an accommodation ladder, a whole flight of stairs which is lowered from a deck. But sometimes he has to climb a Jacob's ladder, which is simply wooden steps fastened to ropes that hang down the ship's side.





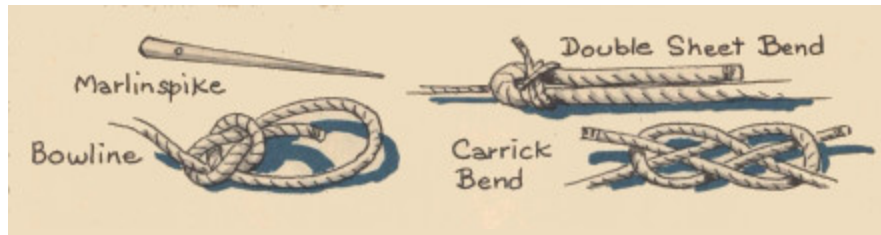
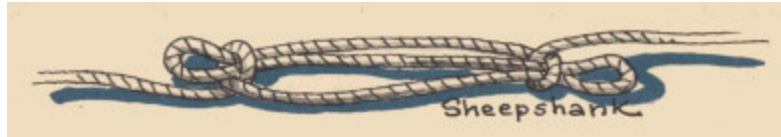
The pictures explain some more words Jim had to learn. A pier or a wharf is a platform sticking out into the water. Ships tie up alongside it. Seamen sometimes call a pier a dock, but a dock is really the water between piers.

A hatch or hatchway is an opening in the deck of a vessel. People can go down a hatch, and so can cargo. Big strong poles called booms raise and lower cargo through hatches. Booms are attached to single masts on some ships; on others, to pairs of posts called king posts or Samson posts or goal posts. When seamen fasten heavy layers of canvas over the hatches, they say they “batten down the hatches.”



Backstay, stay and shroud are all wire ropes that brace the masts. The poop deck is a deck at the stern. Taffrail is the rail around the stern. The taffrail log is a kind of speedometer that tells how far the ship has travelled. It is made up of a line attached to a little propeller which measures miles as it is dragged through the water.

The beam is the widest part of a ship. The keel is the lowest part. The bilge is the low, rounded bottom of the ship. Any water that seeps into a ship collects there and has to be pumped out. Ballast is a weight of some sort, low in a ship to balance her or keep her down in the water so her propellers can work when she has no cargo. Draft is the depth of water needed to float a vessel. When Jim says his ship “draws twelve feet,” he means the keel is twelve feet under water when she is loaded.



OTHER JOBS

A sailor knows how to do many things besides stand lookout and steer. If a line breaks, he can mend it by splicing the ends together with a tool called a marlinspike. If lines wear thin, he puts in new ones—and lines are needed in a great many places on even the most modern ships.

Sailors know how to tie many different kinds of knots. Each one is good for special kinds of work. For instance, a sheepshank is made in a line to shorten it. Jim calls a bad knot a gilligan hitch.

Painting is something else that sailors do all the time. On one trip Jim painted the mizzenmast. For this job he sat in a bosun's chair. You'll see a picture of it on page 31. When he works high above the deck he always has his paint brush tied to his wrist. Then, if it slips out of his hand, it can't fall and hit anyone below.

All the sailors get their orders from the bosun, whom they call "Boats." That's because the real spelling of bosun is boatswain. The bosun gets his orders from the mate on watch who gets his orders from the captain. The captain is in charge of everything. Seamen call him the skipper or the master or the Old Man.



The "Chief" (chief engineer) and his three assistant engineers get orders from the skipper, too. The firemen in the engine room help the engineer carry out the orders. When they are on watch, they look through little peep holes into the oil burning furnaces to make sure the fires are burning just right. They keep an eye on the steam pressure gauges.

At the same time, men called oilers keep every part of the ship's huge engines and other machinery well oiled. On some ships there is a big piston, like the driving rod on railroad engine wheels. One end of it moves in a circle. The oiler has to squirt oil in a little cup at the end of the piston. Every time the cup swings up where he can reach it, he aims his oil can. He is very careful to aim straight. If he misses the cup, oil splashes all over.

No matter how careful he is, some oil does get spilled and spattered around. It is the job of the oiler to wipe it up and to polish all the brass fixtures, which he calls the brightwork. On deck, ordinary seamen polish the brightwork.

One man is in charge of all the food on a ship. He is the steward, and the cooks work under him, and so do the messmen who are the waiters and dish washers.

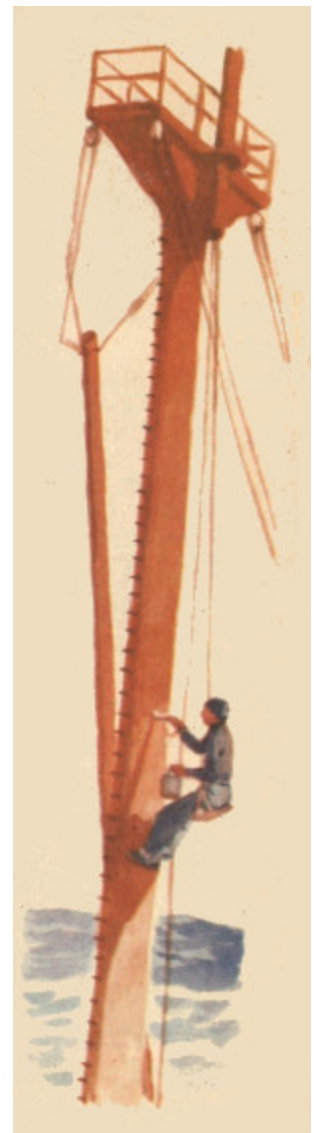
The radio man sends and receives all radio messages. He is called sparks.

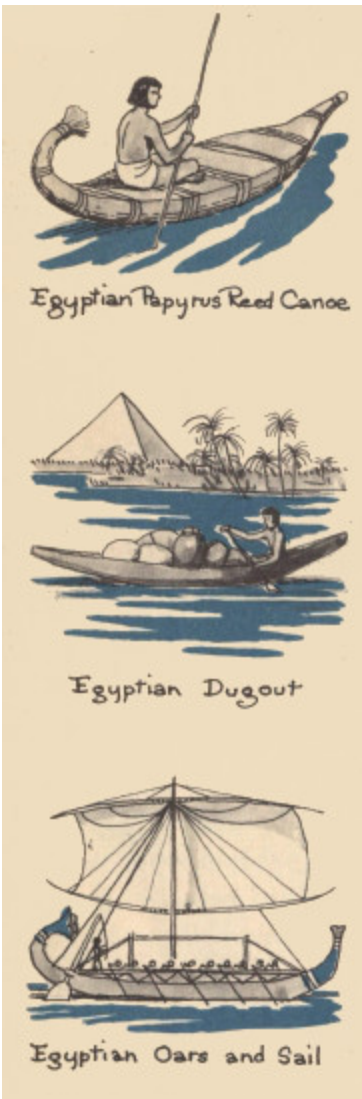
All the seamen who work on cargo vessels, and on passenger vessels, too, are divided up the same way into the deck department, the engine department and the steward's department.

As the great engine deep down in Jim's ship pushes her through the calm blue water of the Mediterranean Sea, he stands watch in the bow. Now he begins to catch sight of small sailing vessels. When his ship enters the port of Alexandria at the mouth of the Nile River in Egypt, he is close to the place where much of the story of ships began.

PAPYRUS REED CANOE. The people of Egypt discovered long ago that bundles of papyrus reed would hold up a man's weight in the water. Later, they tied the bundles into a canoe shape which was easy to handle.

EGYPTIAN DUGOUT. A log hollowed out in the shape of a reed canoe was stronger, and it lasted longer. By adding boards to a dugout along the top of each side, Egyptians had a vessel that could carry





Egyptian Papyrus Reed Canoe

Egyptian Dugout

Egyptian Oars and Sail

bigger loads. Paddles and their own muscles were all they had for power.

EGYPTIAN SAILING VESSEL. Here the power of wind was added to the power of oarsmen. Luckily the winds of Egypt blew from north to south and helped push sailing vessels up the Nile.

GALLEYS. Greeks and Romans used sail-and-oar vessels called galleys. Slaves, chained to their seats, rowed in rhythm. There were many slaves, so their masters could get extra muscle-power by seating two, three or more banks of oarsmen on each side. A ship with two banks was a bireme; with three, a trireme.

DHOW. Other people around the Mediterranean Sea discovered they could do away with oarsmen by making better use of windpower. They invented triangular sails called lateen sails to take the place of square ones. Lateen-rigged dhows are still used. Columbus had both square and lateen sails on the Santa Maria. All three of his ships together were not as long as Jim's freighter.



Greek Trireme



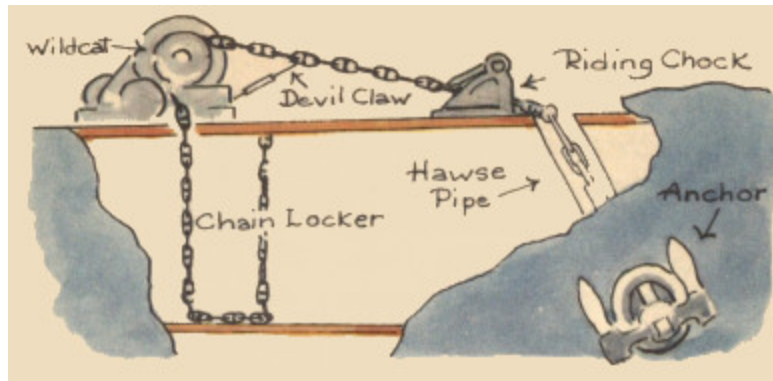
Rowers in a Trireme



Arab Dhow



The Ships of Columbus



New things begin to happen as Jim's ship nears port. He goes down into the forepeak under the deck in the bow. There, all around, are neat coils of hawser which is as thick as his arm. He and other sailors shove one end of a hawser up the ladder. Men on deck grab it and wrap it around a sort of spool called a winch head. Now the winch turns the spool and does the work of lifting out the heavy line. The deckhands lay it neatly on the decks ready to use when the ship ties up at a pier.

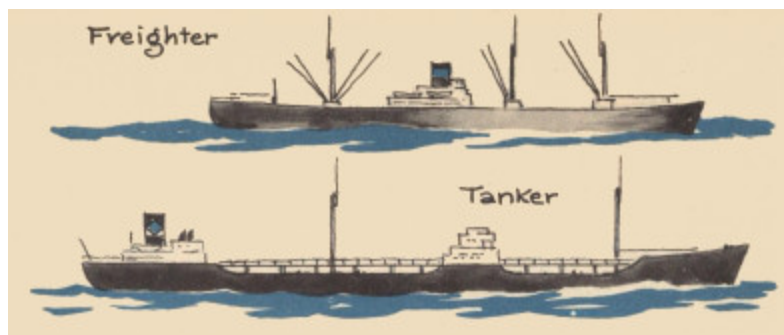
Next Jim goes up to the bow and helps Chips, the carpenter, break cement out of the hawse pipes. A hawse pipe is a hole in the ship's side. An anchor chain runs through it. Whenever a ship raises, or weighs, its anchors and starts on a long trip, Chips plugs up the hawse pipes with cement. This keeps water from splashing up through the pipes in a storm.

On modern ships, a machine called a windlass raises and lowers the anchors. In the old days, when sailors had to raise anchors by turning the capstan by hand, they had a phrase for officers who worked their way up from being deckhands. They said these officers came up "through the hawse pipe." Officers who got their knowledge from going to school and studying books were said to "come in through the cabin window."

After the cement is out of the hawse pipe, Jim takes the devil's claws off the anchor chains and releases the riding pawls. These are two brakes on the anchor chain which you can see in the picture. Now only the brake on the windlass holds the anchor chain in position over the wildcat, which is the wheel on the windlass.



The captain signals from the bridge to let go. Chips releases the windlass brake. The big chain rushes up out of the locker, over the wildcat and down the hawse pipe with a terrific roar. Soon the ship is safely anchored. The skipper can wait now until there is a vacant pier where he can tie up.



After the ship ties up, the captain orders watches broken. The men no longer work four hours and rest eight. Now most of them work eight hours during the day and have the remaining time off, just the way shoreside people do. There is no need for the routine of the sea. Egyptian longshoremen will unload the cargo.

Jim puts on a suit he has kept hanging pressed in his locker. Then he and Juan go down the gangplank. They are off to see the sights in the fascinating Egyptian city—and to buy souvenirs.

But before they have gone very far from the waterfront where a tangle of masts and booms and stacks marks the skyline, they meet Lars, an old shipmate of theirs. That's not so strange as you might think. A sailor often changes ships, and he gets to have many friends who travel just as much as

he does. While they eat an Egyptian meal in an Egyptian restaurant, Lars says he's on a tanker now. She's in Alexandria getting her rudder repaired. It broke in a storm, but the men fixed up something to take its place. They called it a jury rudder.



Lars's tanker looks very different from a freighter. She is long and low and has two houses. One is midships, and the officers' quarters and wheelhouse are there. The crew lives in the other house at the stern.

Between the two houses the deck is so low that waves often wash over it, and so there has to be a high bridge called a walkaway or a catwalk.

Lars says his particular tanker carries "clean" oil. By that he means oil that has been refined into different grades of gasoline. "Dirty" oil is crude oil just the way it comes out of the wells. Lars is a tankerman and a seaman. He has taken a special examination for his job. He knows all the ways to pump different kinds of oil in and out of the tanks on a ship. He knows how to keep gasoline from exploding. He has learned to use special equipment. For instance, he never goes down to clean a tank on his ship without an oxygen mask and a lifeline. The lifeline is tied around him so that a seaman on deck can haul him up if fumes in the tank knock him out.

Like most seamen, Lars has travelled all over the world. In China he has seen junks and sampans. He has seen fishing boats in Portugal with big eyes painted on the bows because sailors thought that helped the boats to see their way. Eyes of the same kind have been painted on ships for hundreds of years in many other places, even in Chesapeake Bay.



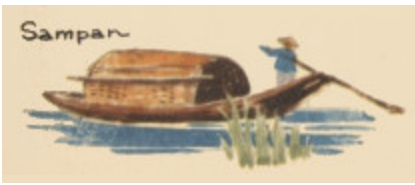
OUTRIGGER. Long ago South Sea Islanders sailed great distances, guiding themselves by the stars. The outrigger at the side gives their small vessel balance in rough water.

JUNK. The sails of this Chinese ship are made of bamboo slats braced by bamboo rods. The rudder is so big that often a dozen men have to work on it. Many junks have colored sails.

WEGIAN SHIPS. Old Viking ships that sailed from Norway had both oars and brightly decorated sails. Vikings were such good seamen they crossed the Atlantic in their open ships. Norwegians are still seafarers. Boys who want to be sailors get training on a sailing ship.

Lars used to work on a tanker that brought oil from the Persian Gulf. When he went ashore there, he saw boats just like the earliest ones that men invented thousands of years ago. He saw boats that were really big, round clay pots, built by people in places where there was plenty of clay but very little wood. He saw huge basket boats woven from a kind of grass and waterproofed with a covering of tar. Some of the basket boats were big enough to carry twenty passengers—or several men and three horses!

Smaller basket boats were used as lighters. (A lighter is any craft that helps to unload freight from another.) Here on the Tigris River, the freight was carried on a large raft supported by animal skins blown up like balloons. A little raft floating downstream sometimes carried its owner, his donkey and the grain he had to sell. After selling the grain, the boatman took the skins from under the raft, let the air out, piled them on the donkey's back and walked back home upriver.



Out at sea, whenever Lars sees a life raft on the top deck, he realizes it is just like the skin-float rafts he saw on the Tigris River. Instead of blown-up skins, water-tight metal containers filled with air hold the life raft up. When Lars puts on his life jacket for lifeboat drill, he is getting ready to use a float, just the way people long ago used bundles of reeds. Even though men have learned so much about ships in all the years since they first started to travel on water, they still use some of the first knowledge they ever acquired.

All of these things interest Lars. He grew up by the sea in Norway, and his people have been seamen since the days of the Vikings. But best of all he likes the clean, modern, comfortable tankers. He is not only going somewhere himself when he is on a tanker. He is also helping to carry a cargo that helps other people to go places.





A Quffa on the Tigris



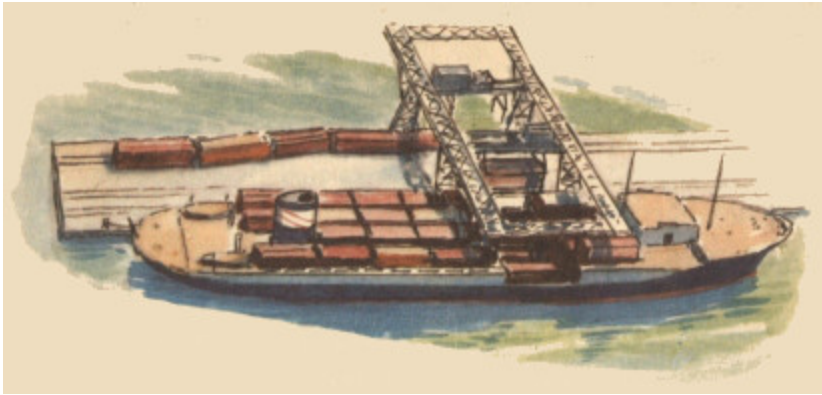
Raft of Timber and Inflated Skins Discharging Grain at Bagdad. Small quffas Serve as Lighters.

Life raft



Life ring





SEATRAINS

Lars's tanker was built to do a very special kind of job. So were many other kinds of ships. Look at the Seatrain, which carries fully loaded freight cars—a hundred of them at a time.

To load a Seatrain, the railroad locomotive pushes a string of cars out onto a long pier. A derrick lifts the cars up one by one, swings them over an open hatch, and lowers them neatly onto tracks in the ship's hold. After the holds are filled, there's still room for more cars on the main deck outside.

It seems queer for trains to travel by ship, but sometimes that's the best way to send cargo. Freight cars can be filled with sugar on the island of Cuba and brought across the water to the United States, without any extra loading and unloading. It's often cheaper for freight cars to go by ship than by rail from New York to Savannah or New Orleans or Texas City.



BANANA BOATS

Banana boats do their own particular kind of work, too. Actually, they aren't boats, although they do carry bananas. They are refrigerator ships. Seamen call them reefers—just as railroad men call a refrigerator car a reefer. Everything about a banana boat is arranged to keep her cargo cool. She is even painted white, because white things reflect some of the sun's rays into the air instead of absorbing their heat. Inside the ship, blowers send cool air circulating around the bananas all the time. It isn't enough just to chill them once and leave them there. Bananas actually make heat themselves. So a constant cool breeze is needed to carry their heat away. The ships that bring bananas from Central America do keep them in the refrigerator.

A banana boat is fast, for she must rush the green fruit from the farm to market as quickly as possible. There are even very quick ways of loading and unloading. Machines called gantries stand on the pier where the ship ties up. The gantries carry the big bunches of bananas in soft canvas pockets arranged in an endless chain. Men on the dock lay the bunches, one after another into the pockets. Men inside the ship take them out and stow them away.

A banana boat sailor does just about the same things that sailors on other cargo vessels do. He steers and stands lookout and works on deck. And like all sailors he has lifeboat drills. Every ship that sails the seas must have lifeboats. Look for them on some high deck, where they are easy to get at in emergencies. Canvas covers on the boats keep out rain and snow and protect the things stowed inside.

A lifeboat is equipped with everything that you may need if you have to float around on the open sea after your ship has gone down. There are water-tight containers full of food, drinking water and matches. There are oars and sails and life jackets, first-aid equipment and ropes. There are flares to light, so that rescuers can locate the boat, and pistols that shoot signal flares like Roman candles high into the air. There are scoops called bailers for dipping water out of the boat. And each lifeboat carries a supply of storm oil. When this oil is spread out on the water, it keeps stormy waves

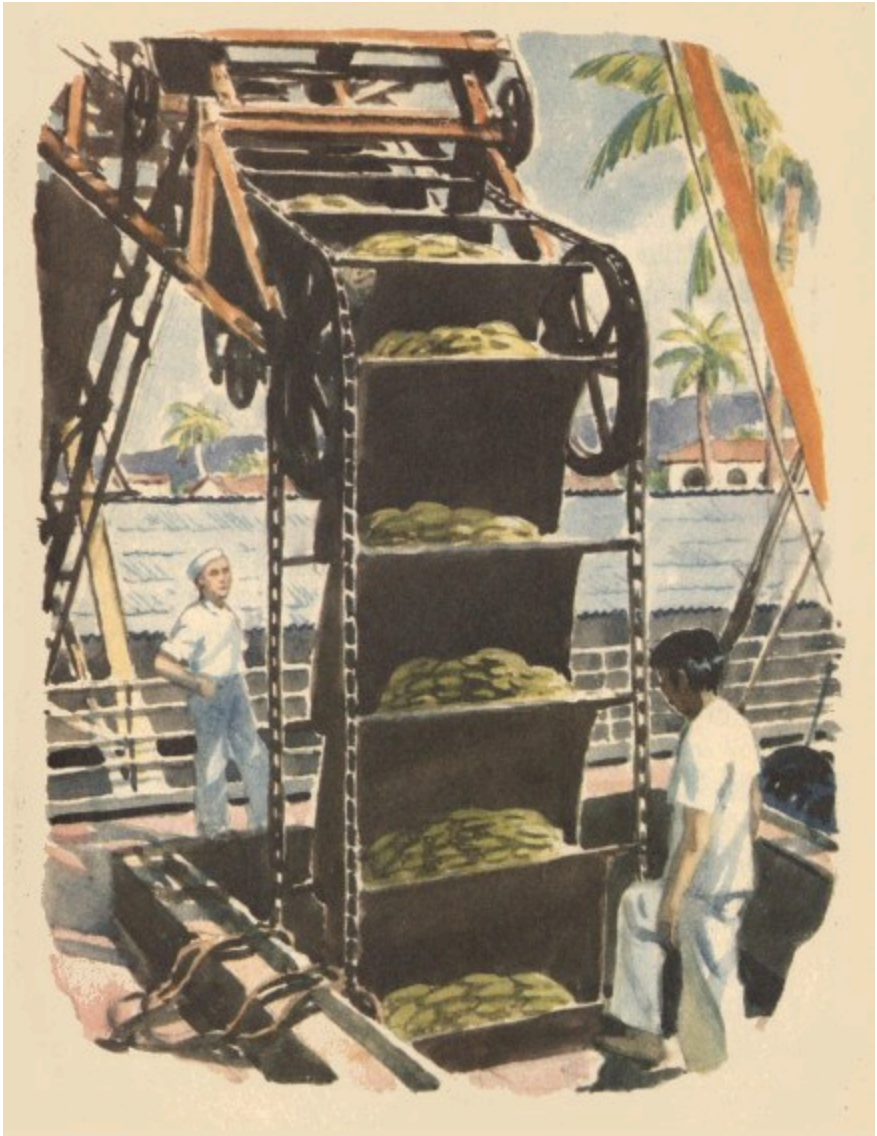
from breaking near the boat. If a wave breaks too close, it may fill the boat with water and sink it.



The can of storm oil fits inside a cone-shaped canvas bag called a sea anchor. The sea anchor floats ahead of the boat and keeps it pointed toward the wind, while the oil drips slowly out and calms the waves. It's important to be pointed into the wind, because a boat that bobs around sidewise can easily be tipped over by a wave. Long ago sailors discovered what a wonderful help oil can be in stormy weather, and that's where the expression "oil on troubled waters" came from. It means to calm things down.

A blast from the ship's whistle tells seamen when it's time for lifeboat drill. Every man knows which boat he's supposed to use. He runs first for his life jacket, then up the ladders by the shortest route to his boat. All the knots and fastenings on the boat are made so that they can be loosened with one jerk. Quickly the men work machines called davits that are always in perfect order, ready to swing the lifeboat out over the water. In a real emergency, the boats would be lowered into the sea, and the men would scramble down rope ladders which are kept ready on deck. But in a drill, seamen just test the davits and lines.

Most lifeboats are double-enders. This means that the bow and stern are rounded and look just alike. The rounded shape helps keep waves from tumbling in at either end. Lifeboats are modeled after the old-time boats in which sailors rowed away from sailing vessels when they went out to harpoon whales.



“THAR SHE BLOWS”

Nowadays, a group of very modern vessels go out together on whaling expeditions. A big ship called the factory ship waits in one place while a half-dozen or more killer boats cruise around hunting whales. The killer boats are power driven, and they are almost as big as an old-time sailing ship.

In June or July, one of these little fleets sets out for the South Pacific. At the whaling grounds, each killer boat begins its search. Suddenly—“Thar she blows!” A whale rises to the surface and spouts. The killer boat dashes after it. The harpooner in the bow aims a gun that’s fastened to the deck. The harpoon in the gun is as tall as a man and heavy, with an explosive charge in its pointed head, and a line attached to the shaft. When the head strikes the whale, the charge goes off inside, killing the great animal. The harpoon barbs spread out. Now the whale is held tight at the end of the line. The killer boat tows it back to the factory ship.

The stern of the factory ship is open. A ramp leads up from the water to the ship’s after deck. Machinery pulls the whale up the ramp and onto the deck. There men with knives that look like big hockey sticks cut up the blubber and throw it into vats where the whale oil is boiled out.

Hour after hour the killer boats bring in whales, sometimes forty or fifty a day—or even more! Everybody works day and night, with very little time to eat and sleep. The oil tanks in the factory ship begin to fill up. Now an ordinary tanker comes alongside. The whale oil is pumped from the factory ship to the tanker which delivers it at some big port thousands of miles away.

When at last the factory ship again has all the oil she can hold, she steams off toward home. For seven or eight months her crew has not been ashore.



Now, as well as in the old days, men on whaling vessels proudly bring home scrimshaw. That is carving they have done on the teeth or jawbones of whales. It is often very delicate and beautiful.

On the return trip the factory ship's speed is much less than when she started out—and not just because her tanks are full. In June her hull was smooth and freshly painted, and it slipped easily through the water. Now in February she has barnacles all over the hull under water—such a rough coat of barnacles that she's held back a great deal.

Barnacles are tiny sea creatures that grow by the millions. They attach themselves to anything under water and form hard little shells. They hold so tightly to the ship that they must be chipped off. That's a job to be done in a place called drydock.

DRYDOCK

All ships go to drydock for regular cleaning and repairing and painting. This is what happens: The ship noses into a place surrounded by three concrete walls. Huge water-tight gates swing shut behind her, penning her in. Mooring lines hold her steady in the exact center of the dock, and pumps go to work taking out all the water in which she floats. Slowly the ship settles into a sort of cradle that has been prepared on the floor of the dock to fit her hull just right. When the water is all out, there she stands, balanced and braced. Now men can work under her and all over her—and inside. They scrape off the barnacles, paint the hull, and repair any parts that have begun to wear out. To reach some parts of the hull painters use long-handled brushes—really long. They're often three times as tall as a man!



Experts go over the ship as carefully as doctors examine people. But many men work at top speed in shifts around the clock, and a ship often spends only twenty-four hours in drydock. Then the gates open. Water flows back into the dock. The ship floats again, ready to go to sea.

Sometimes a ship can't get to drydock. Then a floating drydock comes to the ship. It works the same way as a regular one. Floating drydocks have traveled to distant parts of the world, pulled by seagoing tugs.

TUGS

A tug is a vessel that looks small but has an enormously powerful engine—an engine almost as big as one that moves a cargo ship. In fact, the tugboat's job is to push and pull cargo and passenger ships around.

Big ships need help getting in and out of the narrow spaces between piers in a harbor. If they used only their own power, they might either smash themselves up or crush the piers. Tugs, working together, can push a little here, pull a little there, and ease a huge vessel gently into place.

A tugboat captain must have a great deal of knowledge about the harbor in which he works. In order to pass his captain's examination, he has to draw a map of the harbor from memory, showing every pier and marker and even the rocks, hills and valleys underwater. Most important, he must have a feel for what a ship is going to do when he nudges her at a certain point or when he reverses his propeller and pulls.

For all his skill and responsibility, the captain wouldn't think of wearing a uniform at work. He prefers old work clothes, and he sits down with the crew when the cook serves up jumbo-sized meals.

The cook goes on duty in the galley at any time from one o'clock in the morning on, depending on what time the tug must start work. Breakfast may be at three or four, but the usual time is six. And often the cook's job isn't over at four in the afternoon when he serves supper. If the tug is working overtime, he fixes a meal called a "midnight snack" which the men eat perhaps around seven o'clock. There's enough food in the snack to feed a shoreside person for a whole day.



Besides the captain and the cook, a tug needs a chief engineer, an oiler, a fireman and a deckhand. The deckhand works with the hawsers that are often used when a tug has to pull a big ship.

This is what happens: An AB aboard the ship holds a coil of light line, called a heaving line. At the end of the line is a ball-shaped knot called a monkey fist. The AB gives a big swing and sends the monkey fist and line flying down to the tug. The deckhand on the tug grabs for the line. He's not an outfielder trying to catch the ball. The monkey fist is there only to make the line uncoil and go straight.

The deckhand pulls on the heaving line, which is attached to a hawser on the ship. (Sailors don't say the line is attached or tied. They say it's "bent" to the hawser.) The hawser is so big that it can't be thrown, but it can be hauled onto the tug by the heaving line. The deckhand makes the hawser fast to a bitt on the tug's deck, and now she can pull.

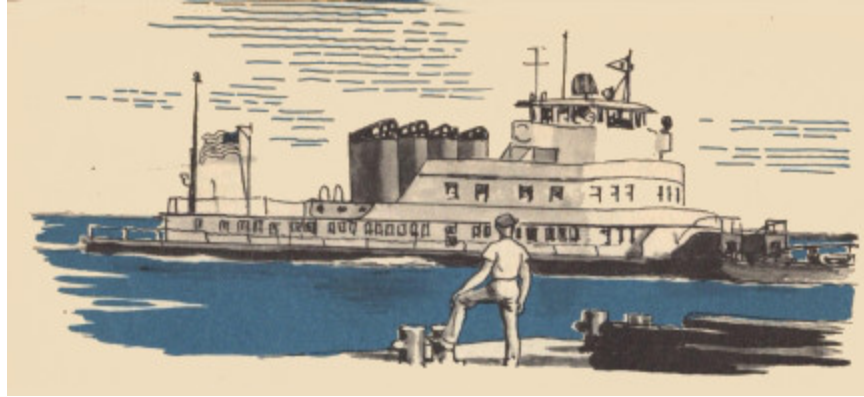
For pushing jobs the tug has a thick pad called a bow fender made of heavy rope hung over the bow. After the fender has been used a while, it gets worn and shaggy and is often called a "beard." It protects any ship the tug is pushing. There are fenders along each side of a tug, too. Sometimes they are made of rope. Sometimes they are old automobile tires or just logs hung loosely over the side. The logs get so much banging around that they may have to be replaced every few days.

Very often a tug has something on its bridge that looks like a gun. It's not. It's a water nozzle attached to a pump, and it's there to help fight fires on ships.

The kind of tug that you can see on the Mississippi River is called a towboat. She doesn't tug, and she



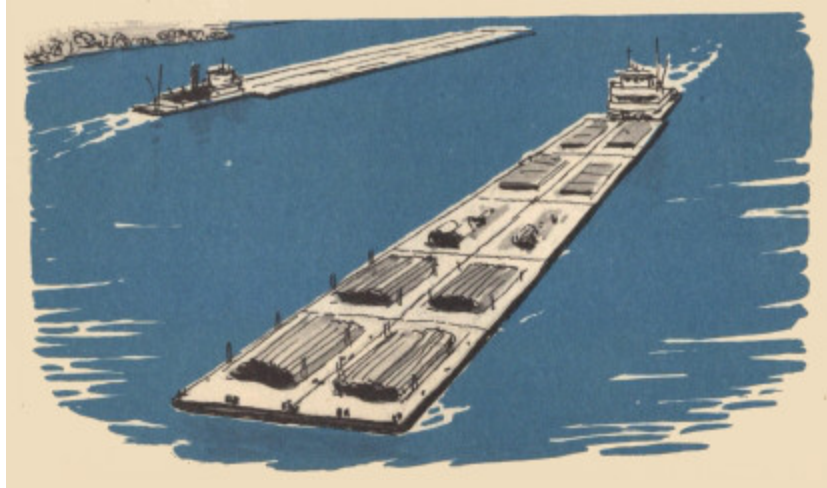
doesn't tow. She just pushes. A Mississippi towboat gets behind a whole string of flat-bottomed barges and shoves them up and down rivers. She often pushes ten barges at a time, loaded with twice as much cargo as an ordinary seagoing freighter can carry.



Many towboats have all of the latest inventions for quick and safe travelling in water that is often more tricky than the open sea. There's a lot of traffic to watch out for on the Mississippi, and the river sweeps around in many bends. Mud collects on the river bottom, so the captain can't always know how deep the water is going to be. Uprooted trees and other big things that could damage vessels often come floating downstream. And when it's pitch dark, or when a thick fog hangs over the water, all these problems get much worse.

Radar is one of the inventions that help towboats avoid danger. Radar sends out radio waves which bounce back to the towboat from anything they hit. In the towboat's pilothouse is a radarscope, which is a little like a television screen. The returning radio waves show up as spots of light called pips on the radarscope. By looking at the pips, the pilot can locate the shores of the river, other vessels, floating trees and anything else that's dangerous.

Another wonderful invention, called a depth recorder, tells the pilot how deep the water is under the head barge in his tow. If the river seems to be getting shallow, he can steer the whole tow into safer water. The depth recorder works by sending out sound waves and making a record of them when they bounce back from the river bottom.



In the old days, river craft had a leadman who measured depth with a line tied to a lead weight. Knots and pieces of leather marked the line. Even at night the leadman could tell by feel how deep the water was. For instance, if his fingers felt that the line was wet up to a place where there were two strips of leather, he would know that two fathoms (twelve feet) of water lay underneath. Two markers at two fathoms. “By the mark twain,” the leadman would call out to the captain.

There was once a Mississippi River pilot named Samuel Clemens who, like all pilots, loved to hear that call. It meant that there was enough water to keep his vessel afloat. Later, when he began to write books, he signed them with the name Mark Twain.

In Mark Twain’s time, the Mississippi River boats were driven by huge paddle wheels. As the wood-burning steam engine turned the wheels, the paddles pushed against the water and shoved the boat forward.

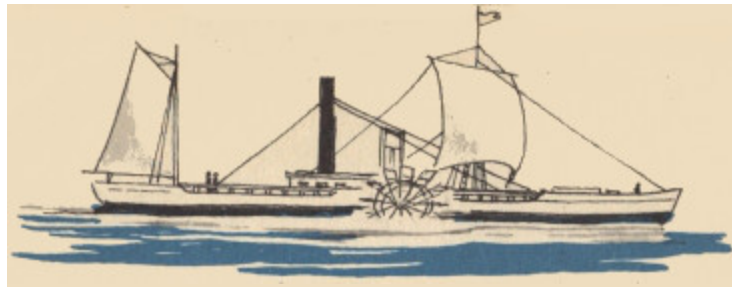
Steam engines began working in rivers very quickly after the first successful paddle boat, the Clermont, proved that she could push upstream. River boatmen needed engines more than seafaring men did, because winds seldom blow upstream as they do on the Nile.

Before there were paddleboats, men took cargo down the Mississippi in keelboats. Then they had to get the boats up-river again almost entirely by muscle-power. Pushing against the bottom with poles, or pulling with ropes from the shore, river boatmen worked the whole way up from New Orleans to Pittsburgh.



A river boatman still works hard, but in a very different way. In his time off, he may listen to radio or even watch television on board the towboat. In the old days, he would have caught fish and fried them over a fire built in a pile of sand on the keelboat deck. Today the cook takes food from a freezer, prepares it on an electric range, and stows the dirty dishes in an automatic dishwasher.

In the old days, the river was the quickest way for passengers to travel, and for freight, too. People now go faster by bus or train or plane. But there's more and more cargo for the barges to carry on the Mississippi and the other rivers that flow into it. Oil, coal, grain, steel, ore, sulfur are some of the things that move along ahead of the powerful streamlined towboats.



GREAT LAKES SHIPS

Grain, coal, ore and limestone for making steel travel on Great Lakes ships, too. So do many other kinds of cargo. Long ago, explorers believed that the enormous sea-like lakes would lead them all the way around the world to China. One man even wore Chinese clothes as he paddled westward in an Indian canoe, so he would be properly dressed when he arrived!

For nearly three hundred years since then, vessels have used these great inland waterways to carry goods and the most precious cargo of all—people. Settlers by the thousand from Germany, Sweden, Scotland and other countries filled the decks of sailing vessels and paddle steamboats that took them right up to the frontier. Today almost five hundred modern cargo vessels shuttle back and forth on the Lakes, carrying the wealth that the descendants of those pioneers have created.

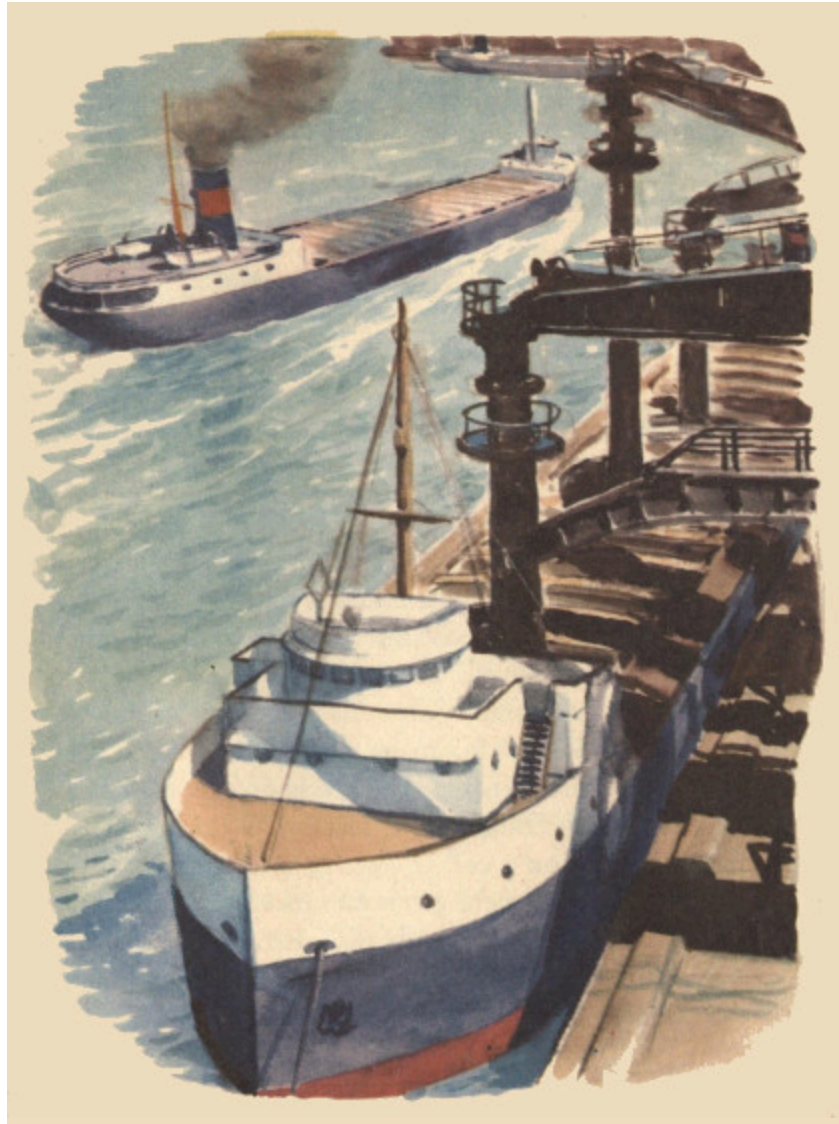
A Great Lakes ship doesn't look like any other. She is broad and low and very long—so long, in fact that she is less rigid than most ships. Seamen say she feels “willowy” if she steams along in heavy weather after her cargo is unloaded. The wheelhouse of a Lakes ship is forward in the bow, along with quarters for the officers and a few passengers. The engine and the crew's quarters are away at the stern. In between, are holds—a great many more of them than on any ocean-going ship. Marvellous loading machines dump ore or any other loose cargo into the holds. Other wonderful unloading machines quickly scoop the cargo out.



Many of the ships run between ports on Lake Huron and Lake Superior. Lake Superior is 22 feet higher than Lake Huron. So ships must use a sort of ladder to get from one to the other through a canal called the Sault Sainte Marie—or Soo for short. Locks in the canal are the ladder-rungs. Suppose a ship is going up. She enters the narrow canal. Ahead are gates. Gates close behind her. She is in a lock. Now the gates in front open and let more water into the lock, lifting the ship higher. She moves forward into another lock and is lifted again in the same way. Sometimes as she goes along, seamen on board toss money to ice cream sellers on shore, and catch the pop-sticks that are thrown back.

For eight months each year, the Lake ships keep hurrying back and forth between Detroit, Cleveland, Buffalo, Chicago, Duluth and other port cities. There's hardly a time when a man can't see smoke from other vessels on the horizon. Then winter comes, and the Lakes freeze over. Lake sailors tie up their ships and go ashore. Most of them have been on the water day and night through the whole season.

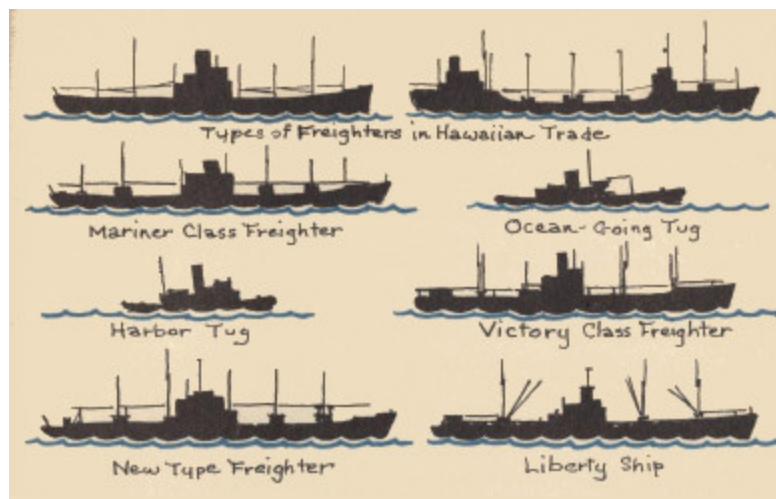
Sometimes a ship stays out too late in the year and can't get to port because ice has locked her in. Then a ship called an ice breaker comes to her rescue. An ice breaker smashes up ice early in the spring, too, so that ships can begin to move.



AMERICAN MERCHANT SHIPS

Merchant seamen man all the different kinds of cargo ships you see in the pictures on these two pages. Their jobs take great skill and patience and very often courage. It has always been that way with men who follow the sea. Some of the things they do are as old as ships themselves. But many things are different now.

On old sailing vessels, the crew had to get their sleep wherever they could find a place to lie down. They might curl up on a coil of rope or on the cargo in the hold. Later, they were given one room, the forecastle, for the whole crew. Everybody was on watch at least twelve hours a day. It is only in the last twenty years that seamen have worked eight regular hours a day.



Almost all ships now have more comfortable bunkrooms, with only two or four men in each one. Instead of living on old-fashioned salt meat and salt fish and crackers called hardtack, seamen have almost the same things that they eat ashore. In the old days, seamen often got a disease called scurvy because they had no fresh food. Then the British discovered that lime juice prevented scurvy, and every one of their ships carried barrels of it. That's why American seamen still call British seamen limeys.

There are laws and regulations now that provide for better food and working hours and pay on ships. Seamen in their unions have worked hard to get the laws and rules that have made life better for them.

FISHING VESSELS

Fishermen have always been among the most daring and hardworking men of the sea. For thousands of years they have experimented and invented, always in search of the boats and ships and nets that will do the best job for them. New England fishermen used to be great whittlers of ship models. They carved out their models partly for fun, partly to give shipbuilders new ideas for improving their designs.

One of the great fishing towns is Gloucester, Massachusetts, and there's a story about it that goes this way: Almost two hundred and fifty years ago, a ship builder in Gloucester launched a vessel that everyone admired. On the day when she first slid into the water, a big crowd gathered to watch. She was graceful and light, and she fairly skimmed along—the way a flat stone does when a boy skips it over the water. In those days in New England, some people called skipping “scooning.”

All at once, someone in the crowd called out, “See how she scoons!” The builder called back, “A scooner let her be!” And according to the story, the name schooner—a new spelling—has stuck to this very day.

A modern schooner still has sails, but not so many as the early ones. An engine now gives her power, so that she can make fast time to and from the fishing grounds, and her sails are used mostly to steady her in the sea while the men work. The engine also helps with the heavy work of handling the nets.



Each kind of fish has its own habits, and the fishermen know them well. Some fish, such as cod and flounder, live down near the floor of the sea. They are caught in drag nets which are towed at the right speed behind the vessel. Men haul the net in, dump the catch into ice-cooled bins in the hold, then drag the net again.

Mackerel behave differently. They swim along in huge groups called schools near the surface of the water. The lookout man on the mast keeps his eye on the sea till he can yell, "School O!" Quickly the men lower a boat that sets a huge net called a purse-seine. At first the net is really a fence. Hundreds of floating corks at the top, and lead weights at the bottom, hold it in place, while the seine-boat draws it into a circle around the fish. Then, at a signal, a motor in the seine-boat pulls on a sort of drawstring in the bottom of the net, closing it and turning it into a kind of giant sack. The seine is "pursed" with the fish trapped inside.

This is what happens on a lucky day. But mackerel can be very irritating fish. Sometimes the whole school will suddenly dive and race away to safety, just the moment before the trap closes. Fishermen must have patience as well as skill.

Before engines went to sea, the men had to purse the seine by hand. Since their schooner carried no ice, they cleaned the fish, salted them and packed them into barrels as fast as possible. Everybody, including the skipper, worked at top speed. Even the cook lent a hand, and he was often a boy of ten who hung his pots in an open fireplace or smoked some of the mackerel in the chimney.



Fleets of fishing vessels go out together when the season is right. There's a race for the fishing grounds, and then a race back to deliver the catch to market. In fishing towns all around the seacoasts, small forests of masts fill the harbors when the fleets are in.

Among the schooners you can also see sturdily-built trawlers, which are usually driven by steam-power. Newest of all are the vessels that work like quick-freeze factories. Machines on board clean the fish, cut them up, package them and freeze them right where they are caught. Or the fishermen may quick-freeze the whole fish, then bring them back to be thawed and sent to market.

People in fishing towns are proud of their fleets, and there's a warm welcome for the vessel that comes in first with a big load.

THE UNITED STATES

The day a ship returns safely has always been important to seafaring men. It's especially important if she has made a new record of some kind. All the seamen in New York harbor were excited when the passenger liner United States came in after crossing the Atlantic faster than any other liner had ever done. And they all showed their respect in the traditional way.

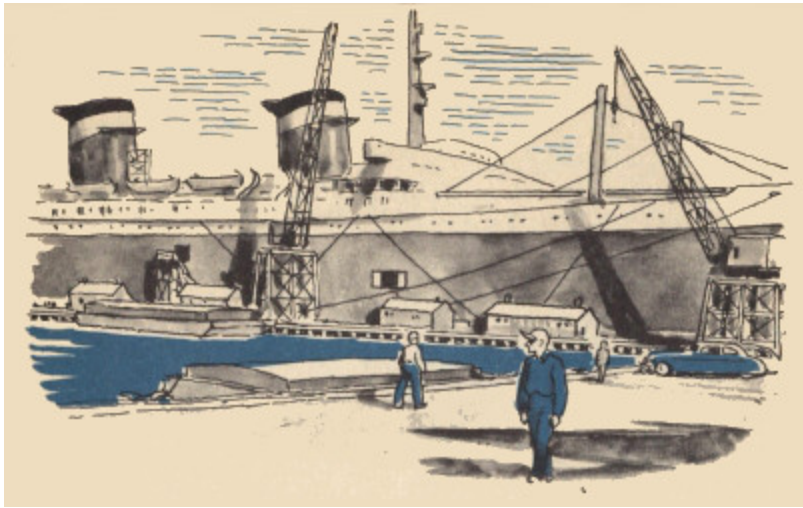
On tugs and freighters, on tankers, on other liners, skippers passed down the word, "Break out the bunting!" This meant take out all the brightly colored signal flags and hang them on the stays. (On page 91 you can find out what the signal flags are.) The United States had her bunting out, too. When she appeared in the harbor, every vessel there greeted her with tremendous whistle blasts. Fireboats filled the air with high curving streams of water from all their nozzles.



Aboard the United States, the members of the crew were more excited than any one else in the harbor, but their work went right on through all the happy hullabaloo. The AB's got ready to tie their huge ship up. Others, from the black gang to the steward's department, were busy with last-minute jobs. Working together as one huge team, they had made the world's fastest crossing. On the trip from New York to England, the United States averaged 35.9 knots. (That means she travelled nearly 42 land miles an hour. Seamen never say "knots per hour." They just say knots.) Before that the passenger liner Queen Mary held the record. It took the United States 10 hours and 2 minutes less than the Queen Mary to cross the ocean.

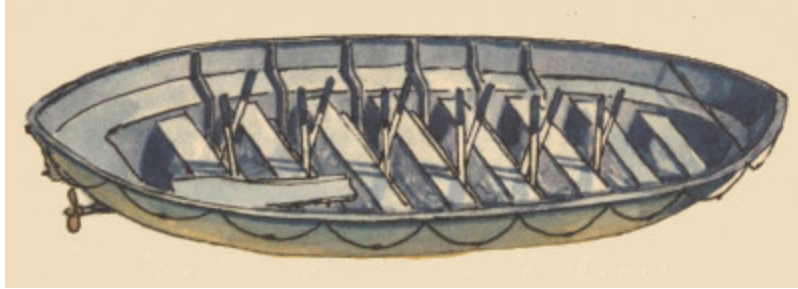
The United States is really more than a ship. With a thousand people in her crew and two thousand passengers, she is a floating town. Besides the

seamen who do their regular seamen's work, there are crew members with special jobs. In the ship's shopping centers, storekeepers sell souvenirs, and all kinds of things that passengers want and need. Movie operators work in her two theaters. A children's nurse takes care of children in the nursery. A veterinarian cares for pets on board. Guards watch over the swimming pool. A doctor and a registered nurse are ready in the ship's hospital to help anyone who is sick. Air conditioning experts see that every room in the ship is kept at the right temperature. Everything from the engine room to the dog kennels is air-conditioned.



Curtains, chair covers and rugs on the ship are made of material that doesn't burn. There is no wood at all in the ship except in the butchers' chopping blocks and in the pianos. But suppose a passenger drops a match into a wastebasket in his stateroom. There's an automatic smoke-smelling gadget that sends a signal to a room on the bridge. The officer there turns on the fire alarm, then pulls a lever which closes that particular stateroom door and blocks the fire off.

There are lifeboats for all three thousand people in case of emergency. These lifeboats are driven by propellers—but they have no engines. People supply the power for the propeller. They push handles back and forth. Even on this most modern ship in the world, there are boats that move in the oldest way—by muscle power.



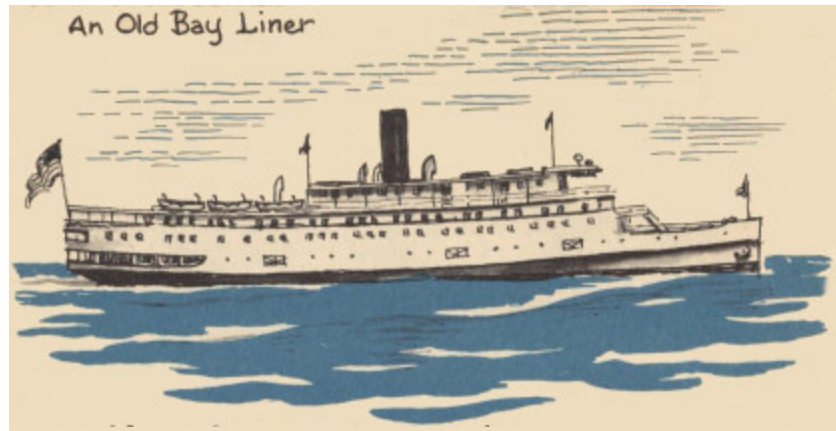
The four propellers of the liner herself are each as tall as a two-storey house. They are turned by enormous steam turbine engines. Smoke from the boilers goes out through unusual-looking stacks. Inside each one are giant filters that take away most of the soot. Besides, there are wings called vanes at the top of the stacks to help keep the smoke from swirling down onto the deck.

Although the United States is about five city blocks long and twelve decks high, she looks as light and graceful in her way as the old clipper ships. The clippers were American sailing vessels that got their names because they went at a very fast clip. A hundred years ago they held speed records all over the world. No wonder the captain of the United States proudly said that his seamen were carrying on the clipper ship tradition.



Many people think the clippers were the most beautiful ships ever built. Certainly they were the first sailing ships to be planned by men who used scientific ideas in their work. At that time, science was bringing modern

machinery of all kinds to the world. Inventors had already put steam engines into ships, but they had not yet studied what was the best shape for a speedy vessel. And speed was becoming very important as more people and cargoes crossed the oceans.



No one knew whether steamships could go fast. But some shipbuilders believed that sailing ships could go faster than ever before. They built the record-breaking clippers. Soon the magnificent vessels began to have races all the way from China to New York and London. It was many years before steamships caught up with the clippers, but in the end they proved to be faster. More important, they could keep going whether there was any wind or not.

OTHER PASSENGER SHIPS

It's the job of a passenger ship to carry people—and give them a good time on their journey. But passenger ships also carry cargo. That's true of big ones and little ones, such as the City of Norfolk which belongs to the Old Bay Line, the oldest American shipping company.

The City of Norfolk goes on short trips back and forth between Norfolk and Baltimore on Chesapeake Bay. She takes on cargo during the day and sails at night. Although she's an old ship, she has radar to help guide her through the busy waters of the Bay. All around are fishing craft, ferries, ocean-going vessels—endless traffic through which the officers must steer a safe course. In the dark wheelhouse, soft small lights hold the key to safety—the sea-green light by which the man at the wheel sees the markings on the compass, the yellow pips and the revolving blue line on the radarscope.

In the hold below are automobiles, piles of second-hand truck tires, crates holding all kinds of things, copper sheets by the ton which have come by train from Utah, and will end up in some eastern factory.

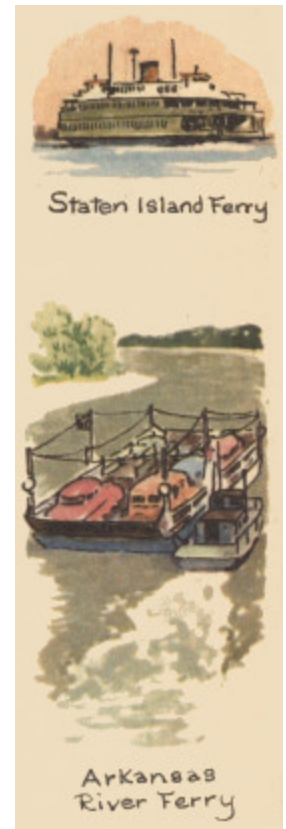


Passengers stroll all over the decks. Some are travelling on business; some are just sailing for fun. A group of school boys and girls on their class trip dance to phonograph records. Their staterooms are air-conditioned, but the inside of the ship looks almost as it did in their grandmothers' day, with balconies and big living-rooms called saloons.

The City of Norfolk—and many other ships like her on bays and rivers and lakes—is really a sort of combination ferry boat and hotel. Most ferries, of course, have much shorter runs, and they are built to fit the needs of their own special work.

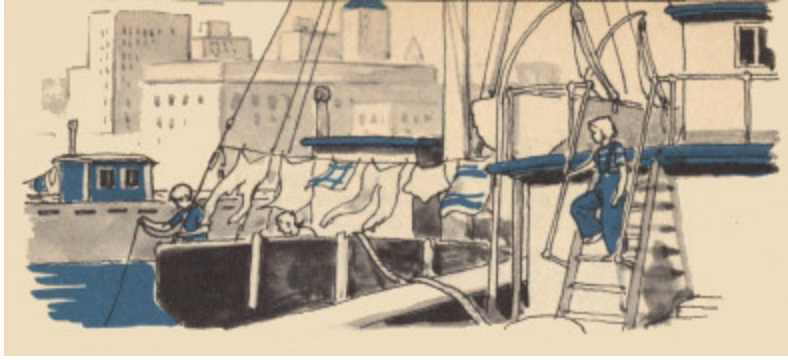
Many ferries look exactly the same fore and aft. They have propellers, rudders and wheelhouses at both ends, and there's a good reason why. A double-ended ferry makes quick trips back and forth. She can save time if she doesn't have to turn around in the water when she goes in and out of her dock which is called a slip.

The big ferries carry automobiles, trucks, and as many as three thousand people at a time. Some of them, on long runs, have up-to-date snack bars so passengers can get quick meals. For safety, they carry lifeboats and life jackets, just as ocean-going vessels do. But a ferry could never go to sea. She is built very broad, with very little of her under the water and a great deal above. Big ocean waves would tip her over.



Men have used ferries from the earliest times. Hundreds and even thousands of years ago people and animals were ferried across rivers on rafts. Even today there are raft-like ferries which men guide across our rivers by steel cables.

Train ferries take loaded freight cars across harbors where there are no railroad bridges. In some harbors, the cars travel on flat-bottomed barges which tugboats shove along.



Long ago, barges were quite different. They were elegant vessels in which kings and important people travelled on rivers. And fancy barges, towed along behind paddle steamboats, once carried passengers up and down the Hudson River, too. At that time the steam boilers on paddleboats often exploded. Many crewmen and passengers were killed. So, in order to attract customers, some steamboats towed “safety barges” behind.

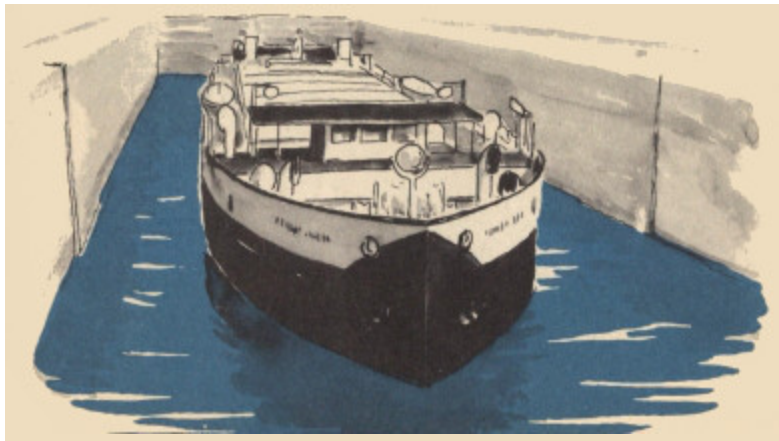
Nowadays barges are plain cargo vessels that do heavy work. Most of them have no power of their own. They must be towed or pushed. The seaman who handles a barge is called a barge captain. He must be an AB to get the job, and on some barges he lives in a house at the stern. If he has a family, they may make their home there the year round.

Before the days of railroads, a whole system of canals joined many of the important American cities. Along these waterways horses or mules pulled barge-loads of freight. Many a canal boatman started before he was twelve years old, driving a mule on long trips all by himself. There are still some canals in use, and powerdriven barges carry cargoes on them.

FIREBOATS AND OTHER HELPERS

The old-fashioned engines that used to explode are gone now. So are the candles and whale-oil lamps that lighted ships. All these caused fires in wooden vessels. But even today, when most ships are made of steel, with fireproofing equipment, there's work for fireboats to do.

The seamen aboard fireboats belong to the Fire Department. They do deck work or engine work, and they also handle the pumps and nozzles that shoot enormous streams of water. The pumps suck in water through holes in the side of the boat and force it through hoses and nozzles that can be aimed like big guns.



Sometimes fireboats go a little way outside their harbor to help a burning ship. On the way, the fireboat captain guides his vessel between buoys that mark the channels where ships can go. All harbors have these channels, which are really streets for water traffic. The buoys are floating signals anchored to the bottom. On a clear day, seamen can tell by looking at the shape and color what each buoy means. In a fog or at night, they listen for the bells or whistles on some special buoys and watch for the flashing lights on others.

Rivers have channels marked with buoys, too, and men who belong to the United States Coast Guard Service have the job of placing and repairing them.

The Coast Guard also cares for lighthouses at dangerous points along the shore. Powerful lights and foghorns in the lighthouses warn ships away

from rocks or shallow water and also help them find out exactly where they are. In some places, lightships anchored in the sea do this same job. A lightship is really a giant buoy. Seamen live aboard her to care for the safety equipment. They get their food and mail from vessels called tenders. (Any vessel that supplies another is a tender.)

Coast Guardsmen help seamen in other ways, too. Suppose a ship is sinking. Fast, tough little Coast Guard cutters race off to the rescue the minute the dreaded SOS signal comes over their radio. (SOS is the code



signal for “help!” and every radio man understands it, no matter what language he speaks.)



Using a special gun, men on the cutter shoot a lifeline across to the sinking ship, and a breeches buoy is rigged on it. This is a canvas seat, made like a pair of short pants. The seat hangs from a wheel called a block which runs along the line. One by one the seamen sit in the seat and are pulled along to safety.

CHARTS FOR SAFETY

In the days when the United States was still a very new country, many people in Europe longed for the freedom they were sure they could find here. One of them was Ferdinand Hassler, a young Swiss mathematician. Hassler was no seaman when he set out for the new world in a sailing ship. But luckily he did know a great deal about the stars. After the captain of his vessel collapsed in a terrific storm, Hassler was able to look at the stars and tell the seamen how to steer the ship.



The things Hassler knew about mathematics made it easy for him to navigate, but real troubles began when the ship came into Delaware Bay. The map of the bay was old and very inaccurate. Hassler could not tell whether the ship was in shallow water or deep water, except by watching the leadline day and night.

This last part of his adventure made young Hassler very angry because it was so unscientific. He realized that the safety of all ships depended on accurate maps, called charts, of the coasts and harbors. Soon after he landed he began to make plans for a survey of the whole American coast. He talked to President Jefferson who agreed with him, and Congress finally gave him the job. At last his good charts began to help save lives.

Today the United States Coast and Geodetic Survey carries on the work Hassler started. Using many ships and small boats with marvellous equipment, the scientific men of the sea go about their important and often dangerous work. It's their job to map the earth that lies under the oceans, rivers and harbors. Here are some of the things they do along the Alaskan coast.

A survey ship moves through the water sending sound waves to the bottom of the ocean. A delicate machine records the echoes made when the sound waves bounce back from the deep valleys and the high mountain tops that lie beneath the surface. Scientists know how fast sound travels in water, so they can tell exactly how deep it is. But some of the peaks are narrow and sharp. Even the wonderful machines miss them. And so these very modern vessels must do something old-fashioned and simple. Two of them travel side by side with a long wire cable hanging between them. If the cable catches on a rock, the men know they have snagged a sunken mountain top.

Often, men on shore help the men on ships with their surveying. The instruments they use are so delicate that the warmth of direct sunlight would cause inaccuracies. One slight error might mean a shipwreck. So, even in Alaska, a surveyor works under an umbrella.



You might think that all the charts and maps should have been finished in the long years since President Jefferson's time. But the work of making charts can never be finished. The coastline is always changing. Currents and tides, storms and floods shift millions of tons of sand near the coast. Earthquakes and volcanoes raise land or lower it. A place that was safe for ships yesterday may be dangerous today.

In Alaska, glaciers that run into the sea grow bigger or melt back. Sometimes these enormous rivers of ice push themselves out under the water. The little survey vessels mapping the ocean's bottom have to sail over the sunken glaciers. There is always danger that, at any moment, a great mountain of fresh-water ice may break loose and rush toward the surface. When this happens, any vessel nearby is almost certain to be destroyed.

Seafaring men need to know about the tides when they enter or leave harbors. Tides are very different at different places along our enormous coastline, and they change from one day to another. The Coast and Geodetic Survey has worked out a wonderful way of telling ships about tides in advance. Every day, records pour into Washington from all along the seacoast. The figures they give are put into a fantastic "thinking machine," together with other figures about the sun and moon which cause the tides. Electricity is turned on, and in no time the machine tells what tides will be like with ordinary weather tomorrow or even next month.

Men have come a long way since they first learned to float on a blown-up animal skin or a bundle of reeds. For thousands of years, they have been inventing new and better ways to travel across water. But the oceans have an enormous power and force. Science and seamen still have much to learn about the power which they must fight and make work for them—and which will always be exciting.



WHAT SEAMEN SAY

Here are some words that you haven't met in the rest of the book. They are all part of seagoing language.

AHOY—a call given by men on one ship to greet men on another.

AVAST—an officer shouts “avast” if he wants a seaman to stop hauling on a line.

BELAY—to tie or make fast. A belaying pin is a short rod which can be stuck into a holder so that a line can be twisted around it. There were many belaying pins on old sailing vessels, and they made handy weapons at times.

DEEP SIX—when a sailor throws something overboard, he “gives it the deep six.” The expression comes from the days when sailors measured the depth of water with a leadline. The “deep six” was a place on the line which showed the water was six fathoms (36 feet) deep.

FOUL—Seamen use this word to describe anything that has gone wrong or got mixed up. A snarled line is foul. A ship's hull covered with barnacles is foul. Bad weather is foul.

NORWEGIAN STEAM—seamen say they use “Norwegian steam” when they do heavy work without the help of machinery.

SCUTTLE BUTT—the drinking fountain on a ship. Because seamen often gather there to talk, the rumors and gossip that they pass on are also called “scuttle butt.”

SEA LAWYER—a seaman who likes to argue about rules and regulations.

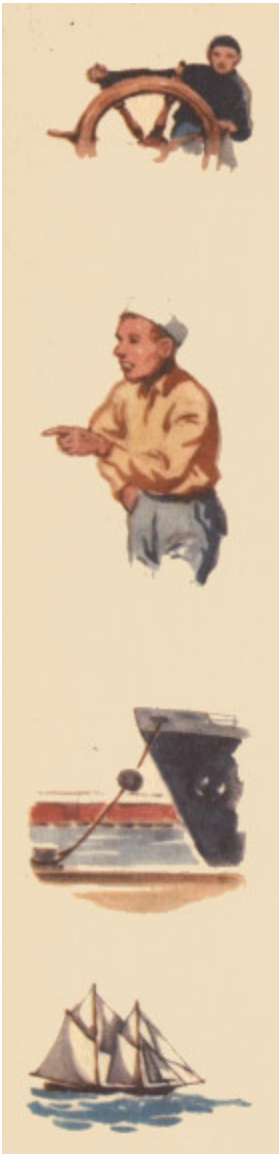
SLOP CHEST—a room where seamen can buy clothes. Every ship is required to have one.

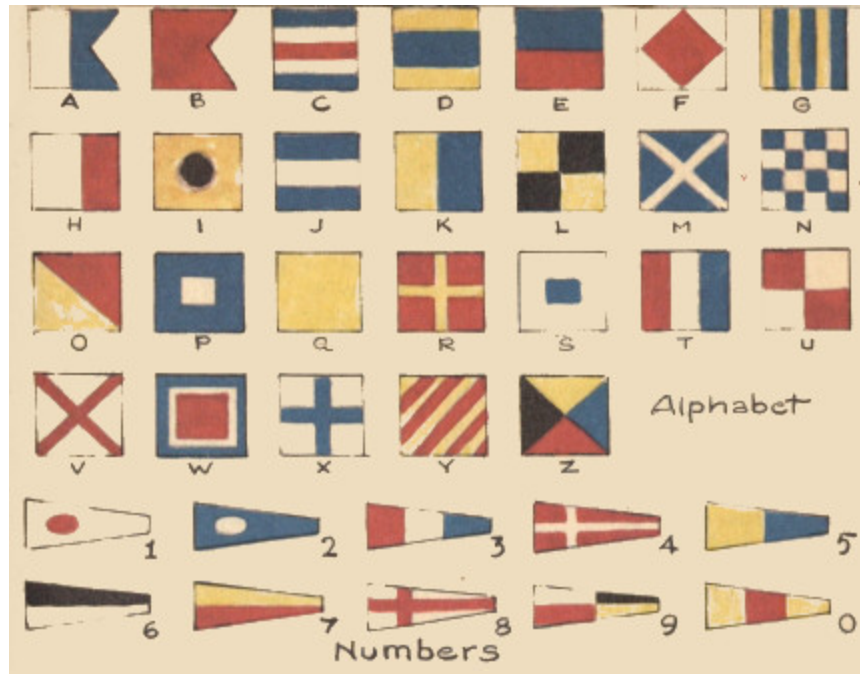
SLUMGULLION—a seaman's word for stew that he doesn't like.

TRAMP—a freighter that ties up anywhere and has no regular schedule.



WINDJAMMER—a sailing vessel.





Code flags make it possible for ships to talk to each other at sea. Each flag stands for a number or for a letter in the alphabet. The flags are used in combinations—not to spell out individual words, but to send a whole message. For instance, the two flags N and C flown together mean, “In distress. Need prompt aid.” No matter what language a seaman speaks he knows what this signal means. Some of the other messages he can read are IQ—“Do not pass ahead of me”; RW—“Where are you from?”; AG—“Shall not abandon my vessel.”

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SHIPS AT WORK

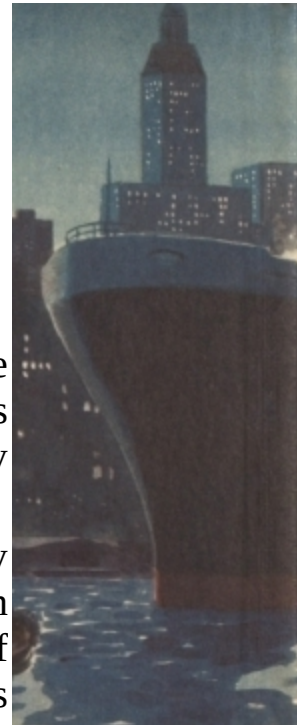
By Mary Elting

Illustrated by Manning deV. Lee

Here is the colorful, exciting life of the sea—the men, the ships they sail, the work they do, the cargoes they carry to the far corners of the world—all vividly presented.

Freighters, tankers, ferries, tugs, and the many unusual ships that do highly specialized jobs are shown in action. The work, the sailor's language, the kind of life a seaman lives, the use of recent inventions (such as radar) all contribute to this fascinating picture of SHIPS AT WORK. The newest and proudest of ocean liners, the "United States," is pictured and described as well as the humblest dugouts and sailing vessels of ancient times.

The illustrator, famous for his marine paintings, has combined beauty with clear, sharp detail. His many full-color pictures in this book give added interest to your seafaring knowledge.



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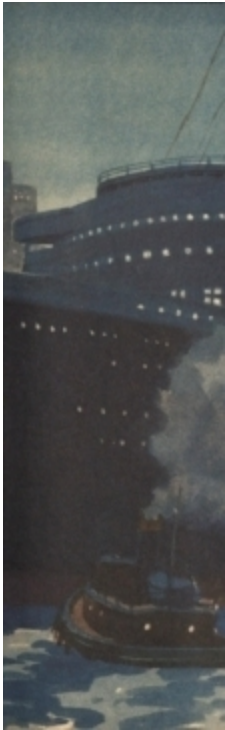
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You will find this book an exciting companion to
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MACHINES AT WORK

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There are machines to dig, to hammer, to push—to do every kind of heavy job and to make work thousands of times easier and faster.

On farms, in the mines, in cities where huge buildings are built and out in the woods where powerdriven saws slice through great trees, many kinds of special machines do many kinds of remarkable jobs.

Can you imagine a giant shovel so huge that it took 45 freight cars to haul it from factory to mine? Do you know that there is a machine that plucks the feathers off chickens, ones that pick corn, dig potatoes? Inventors of machines work on everything—they even had fun making a mechanical mouse that can sniff about until it finds a piece of “cheese” and then “remember” and run straight to it next time!

As marvelous and complicated as all these machines are, the author points out that no inventions will ever be as wonderful as the men who invented them—and the men who make them work.

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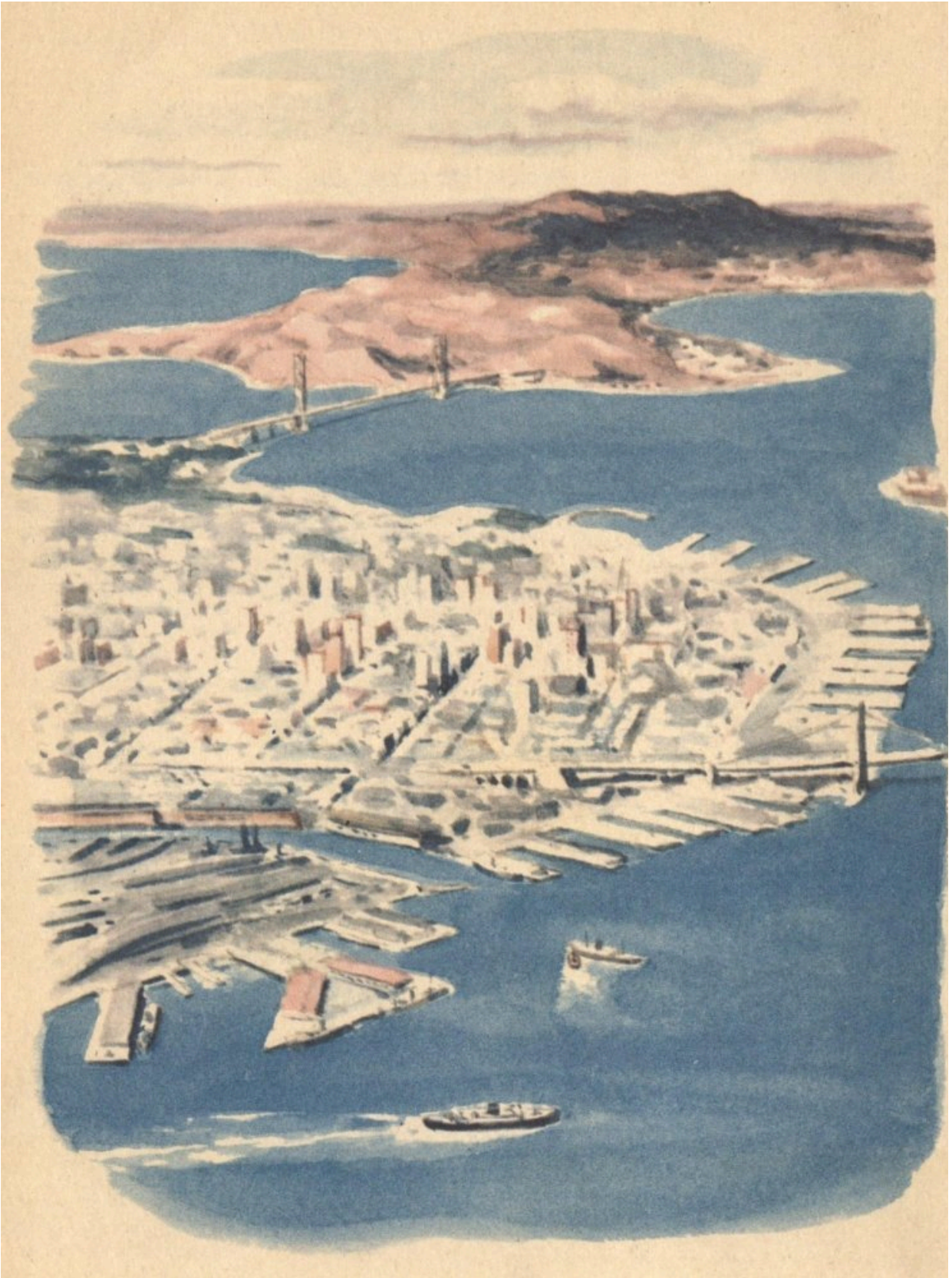
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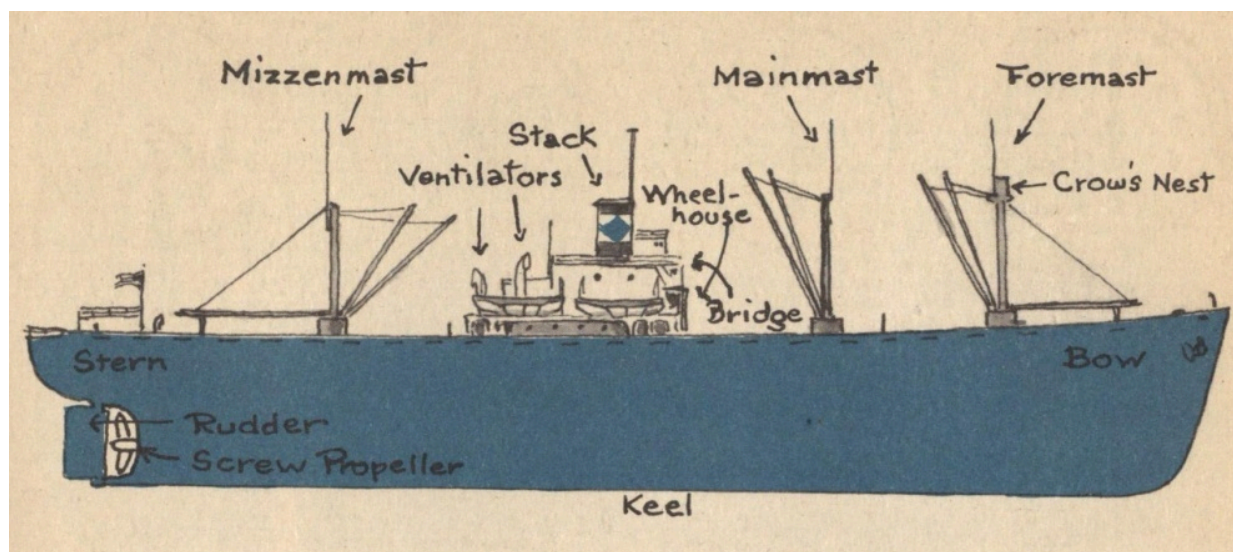
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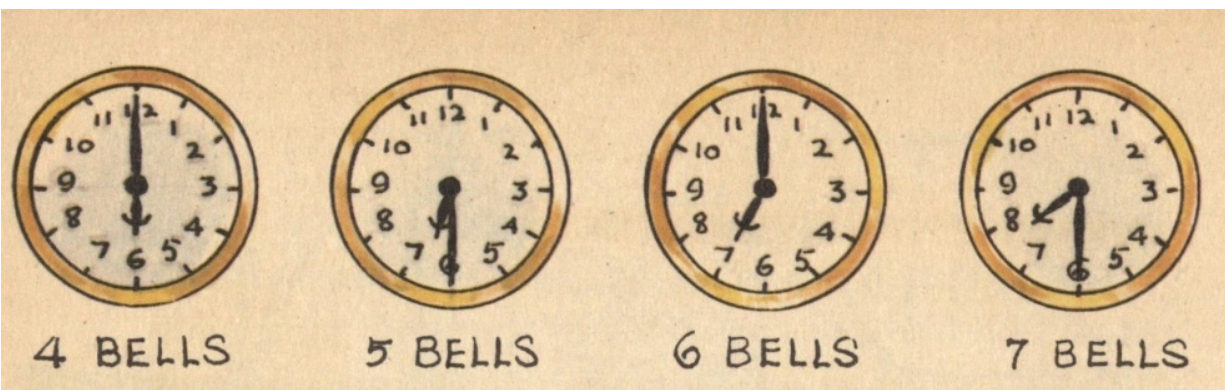
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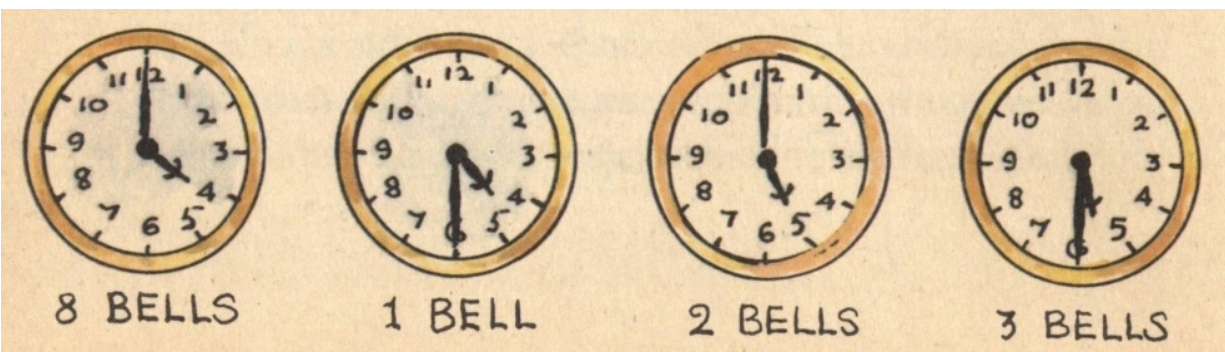
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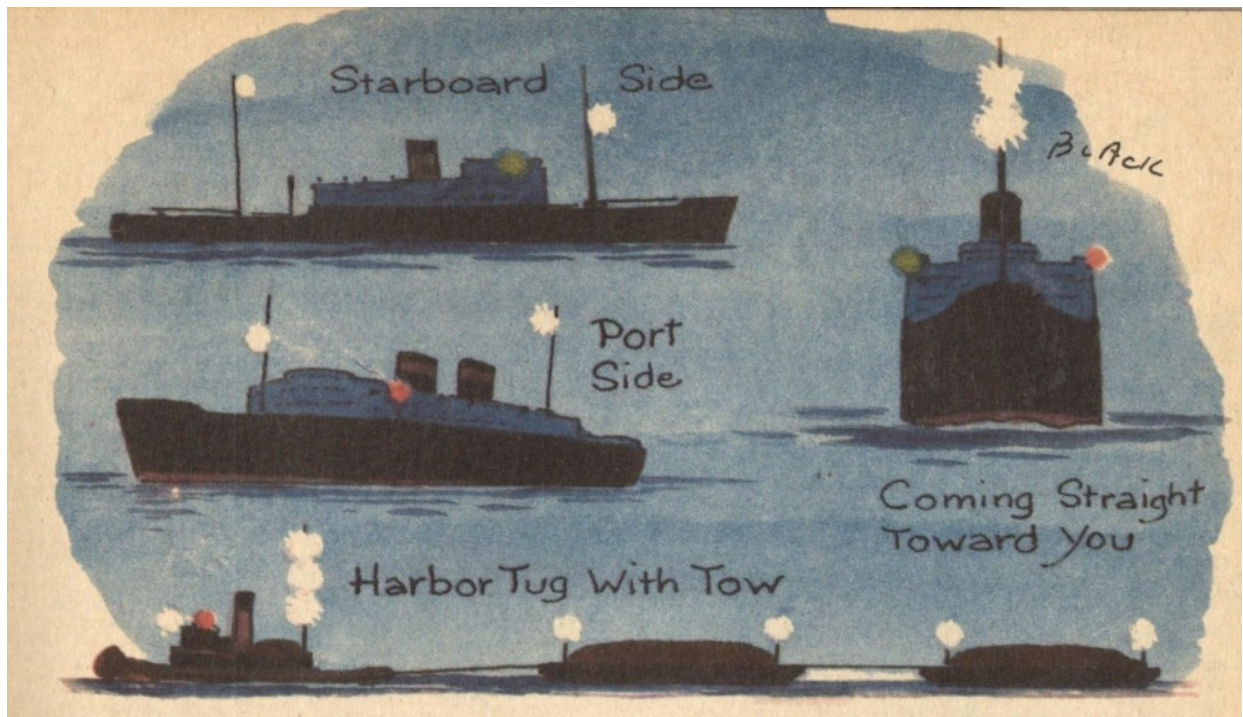
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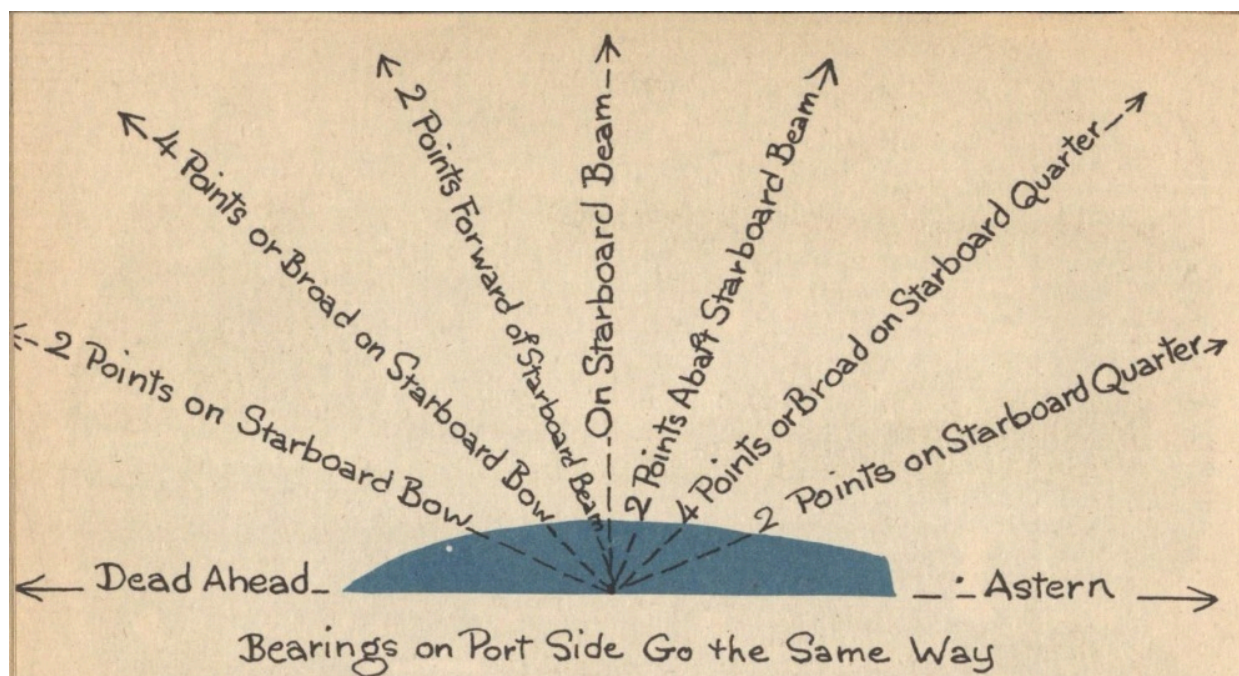
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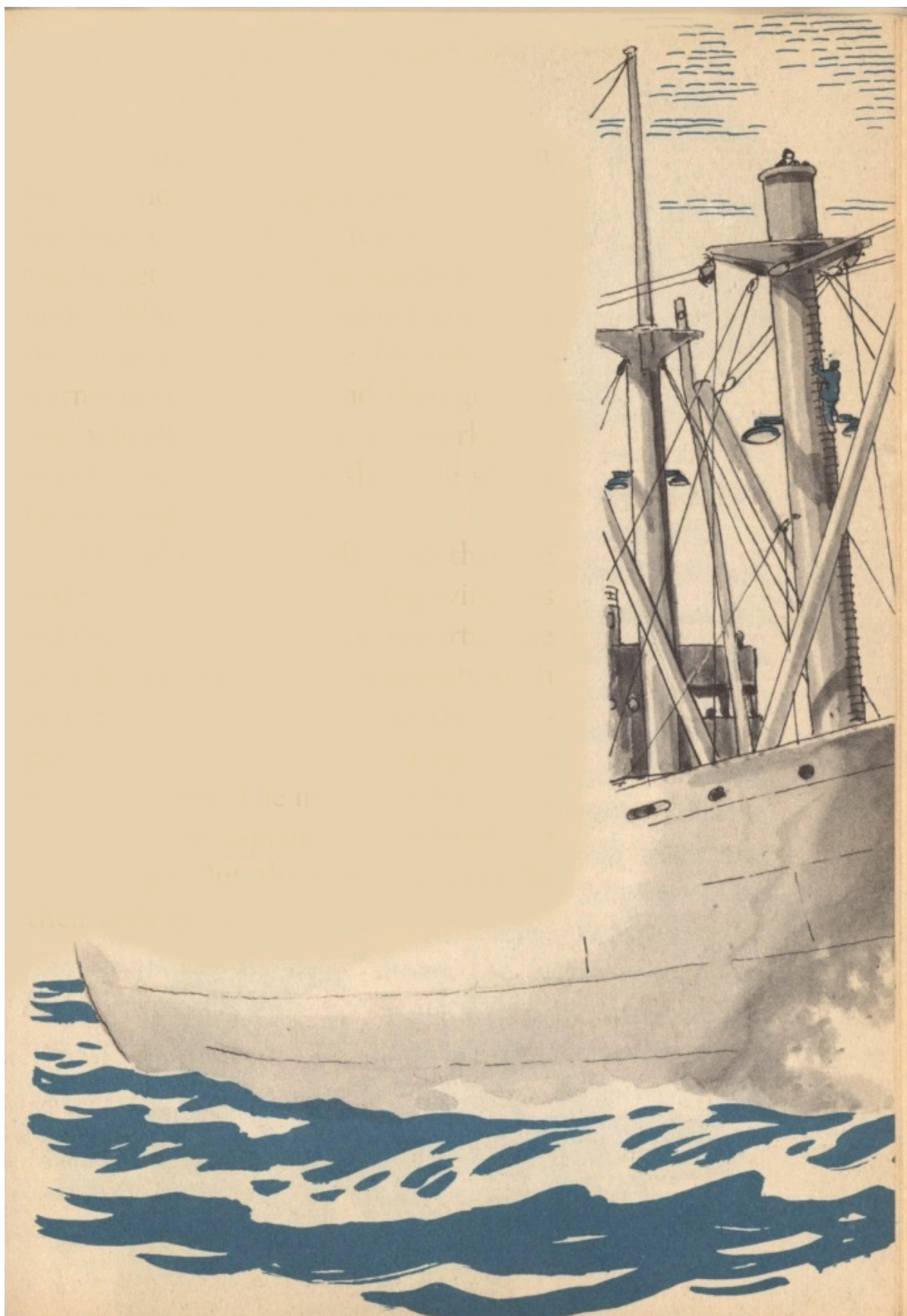
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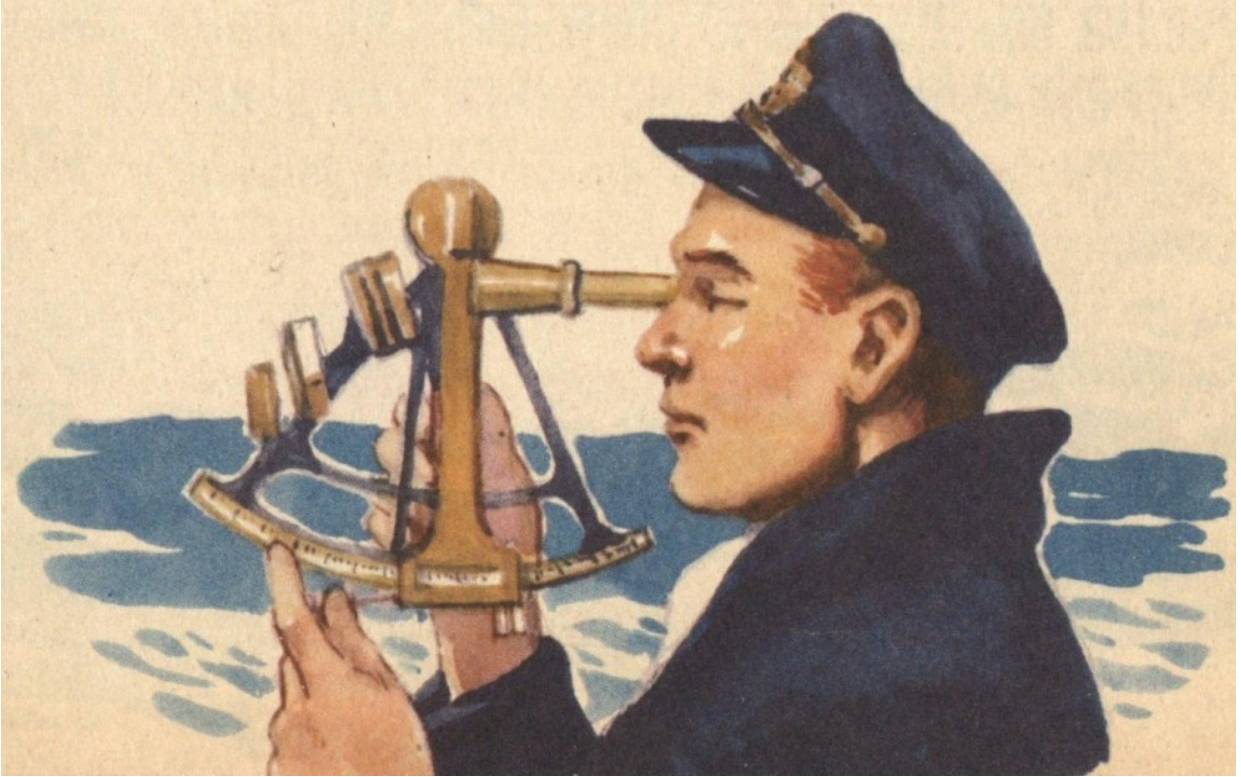
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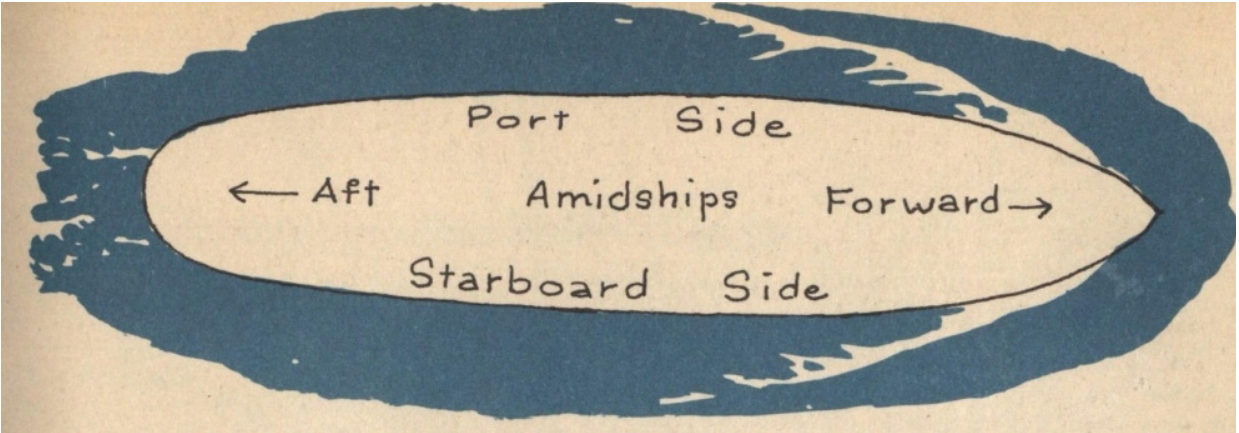
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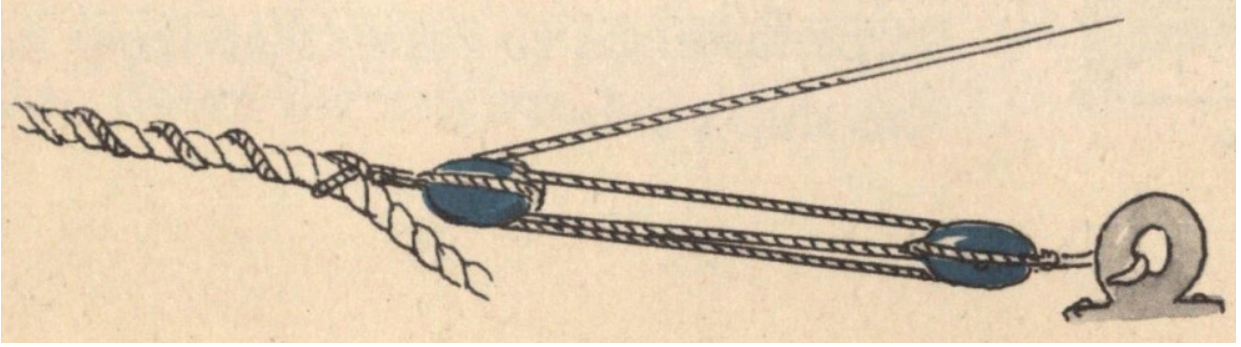
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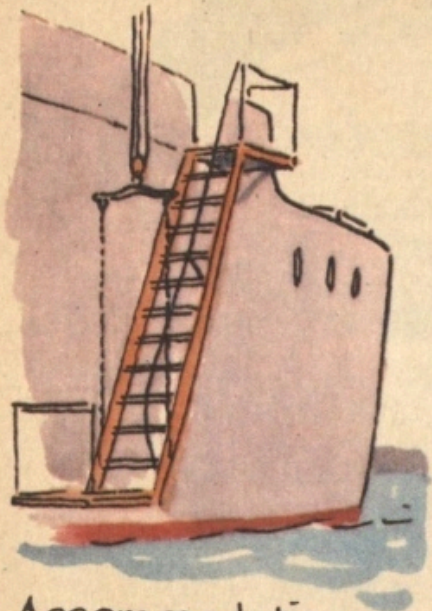
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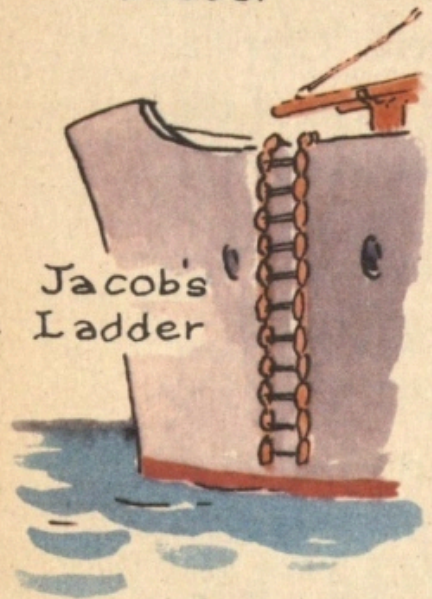
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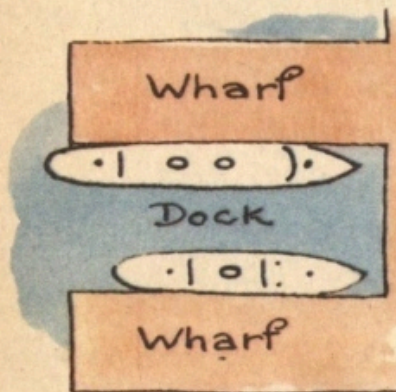
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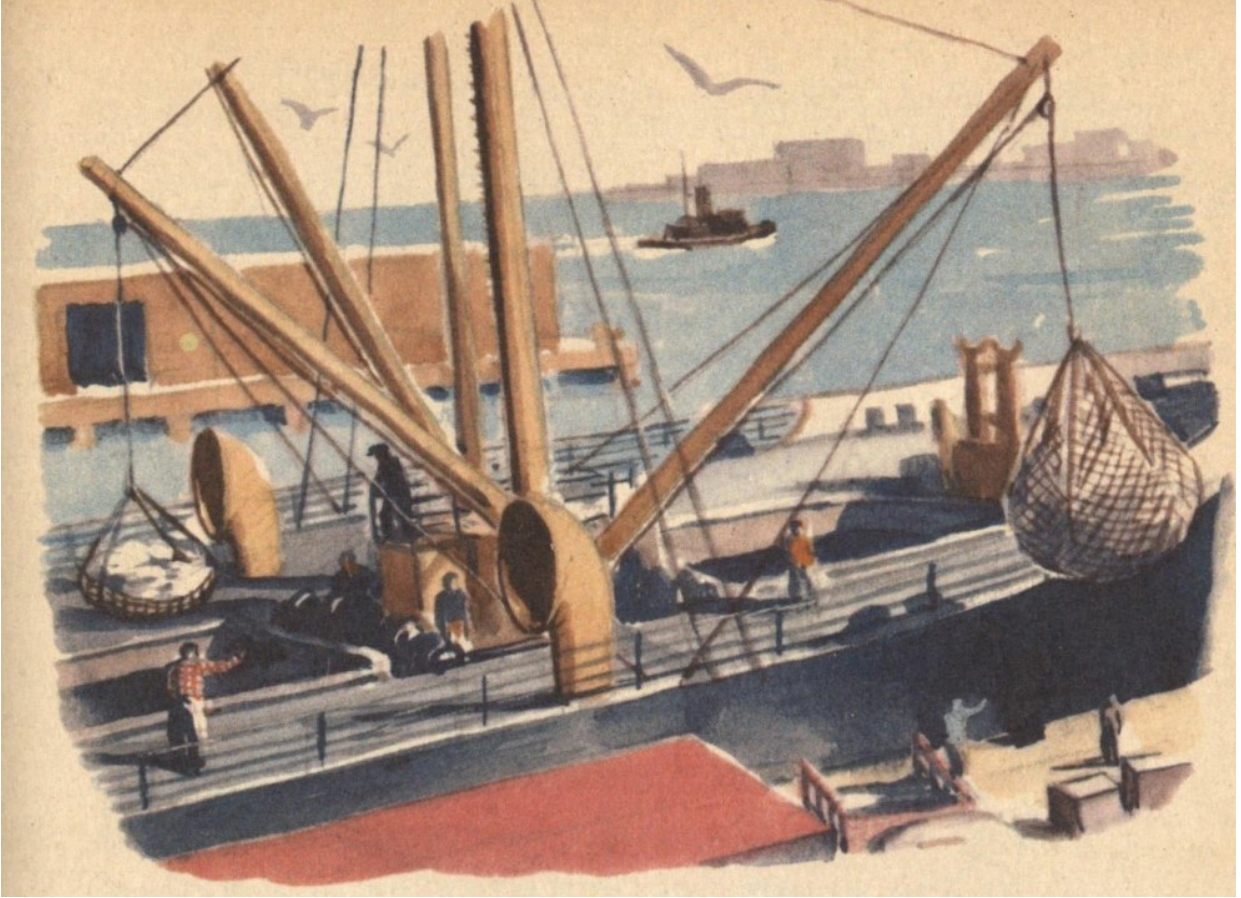
Accommodation
Ladder



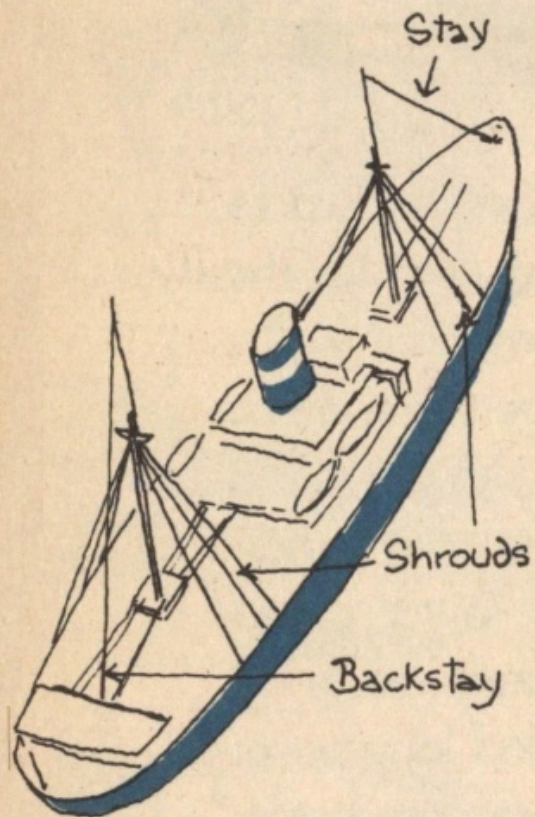
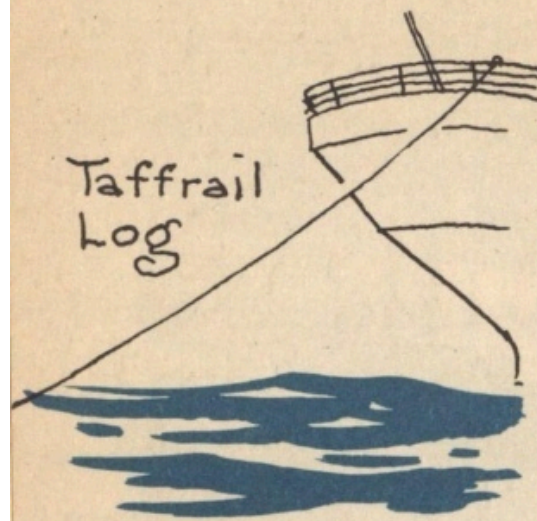
Jacob's
Ladder



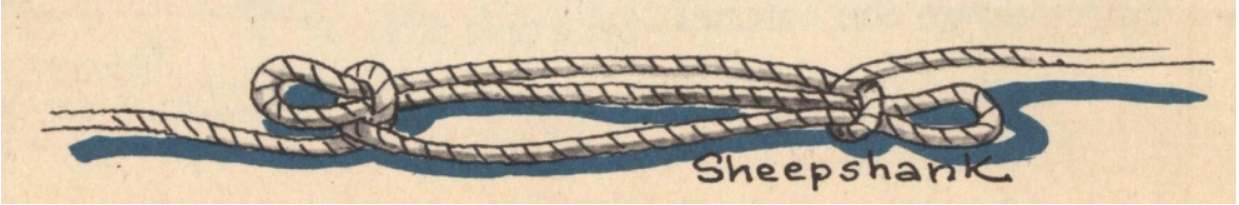
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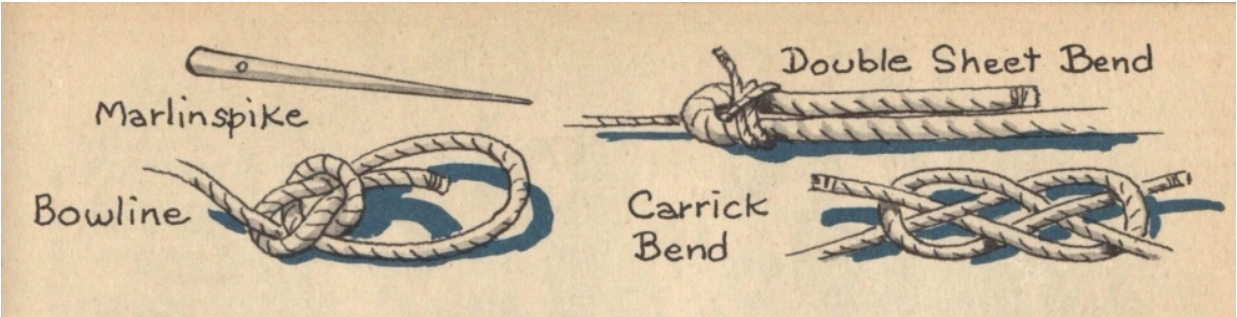
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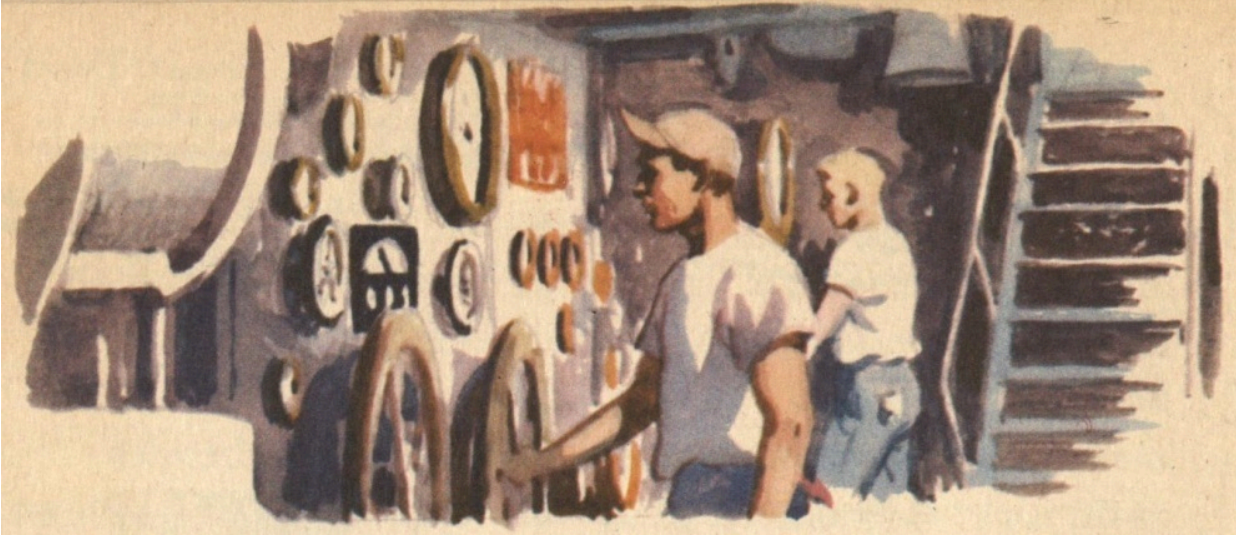
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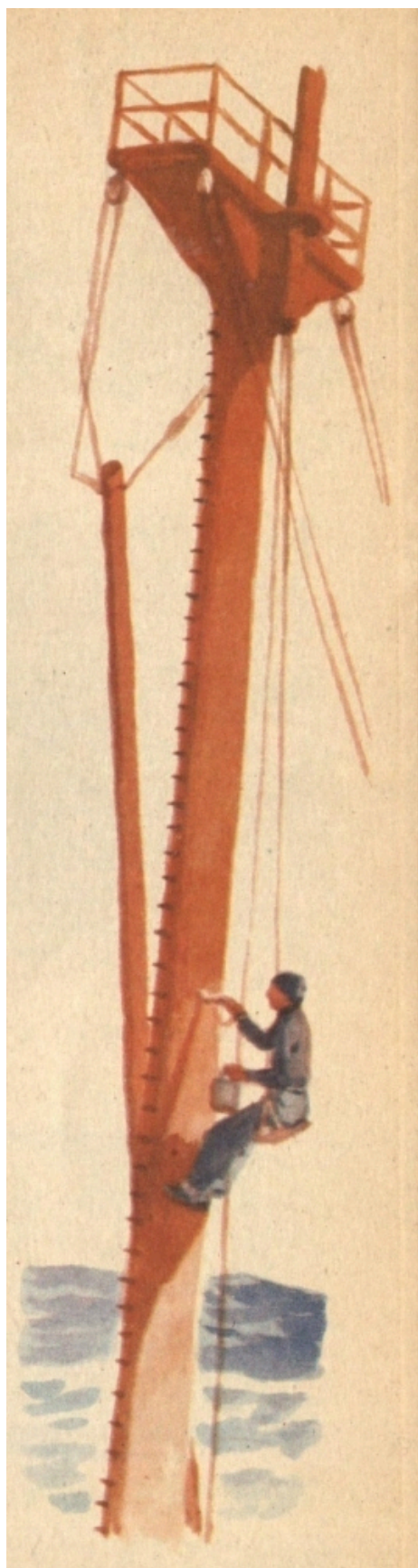
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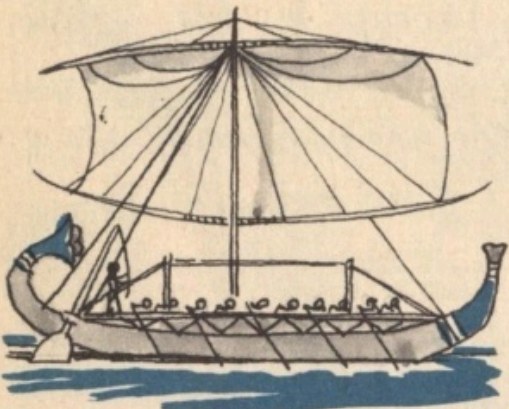
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Egyptian Papyrus Reed Canoe

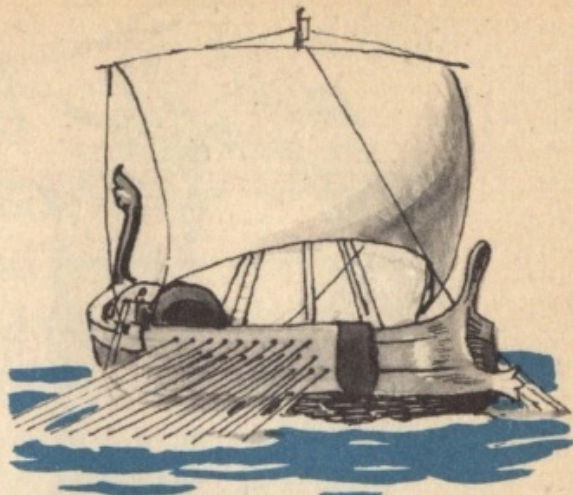


Egyptian Dugout



Egyptian Oars and Sail

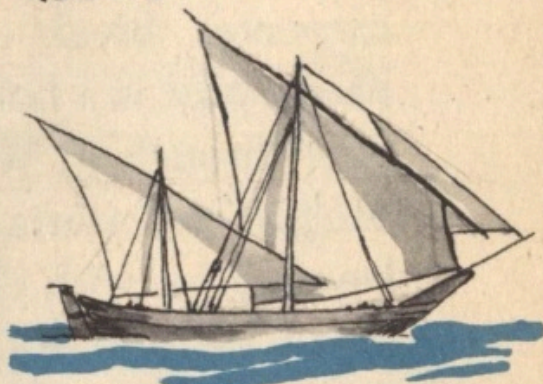
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Greek Trireme

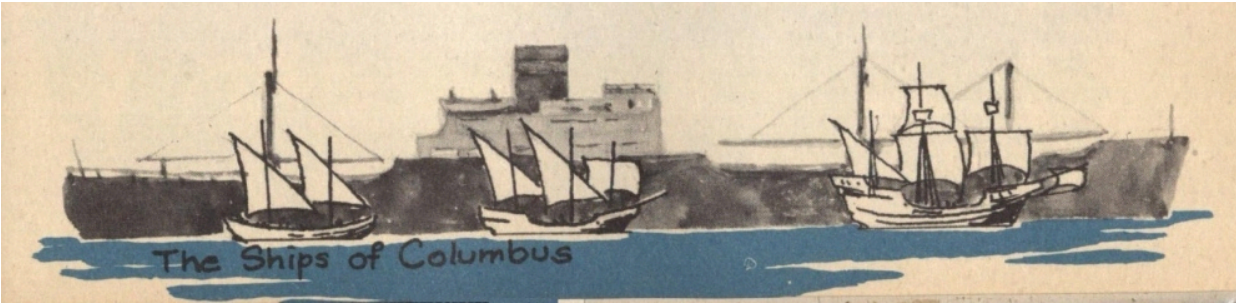


Rowers in a Trireme

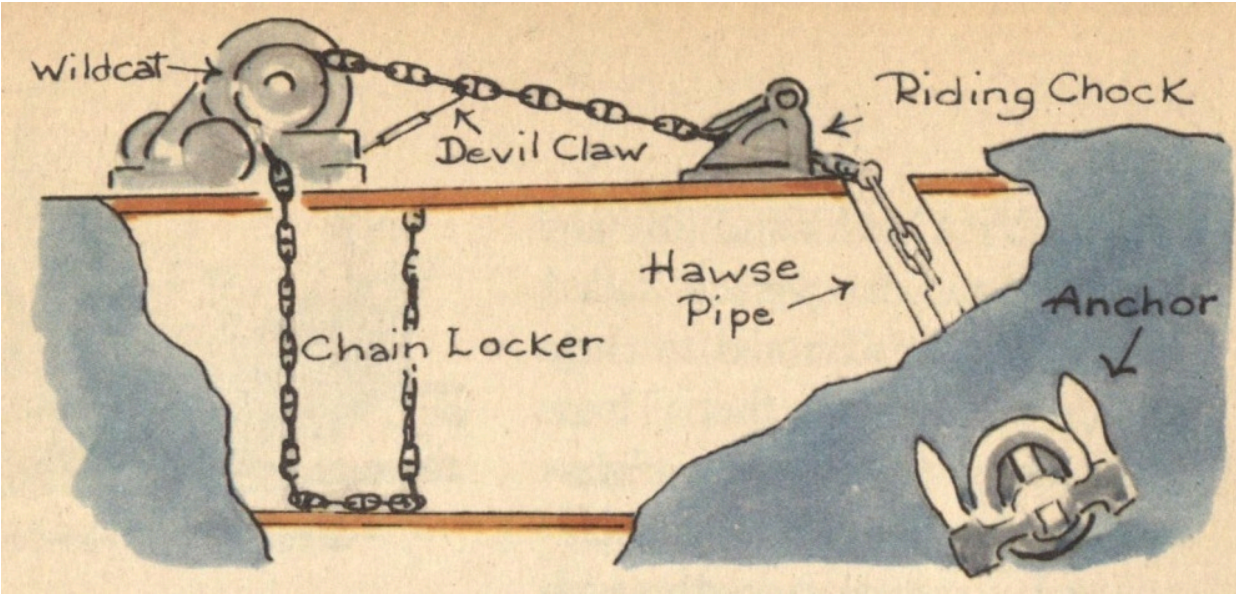


Arab Dhow

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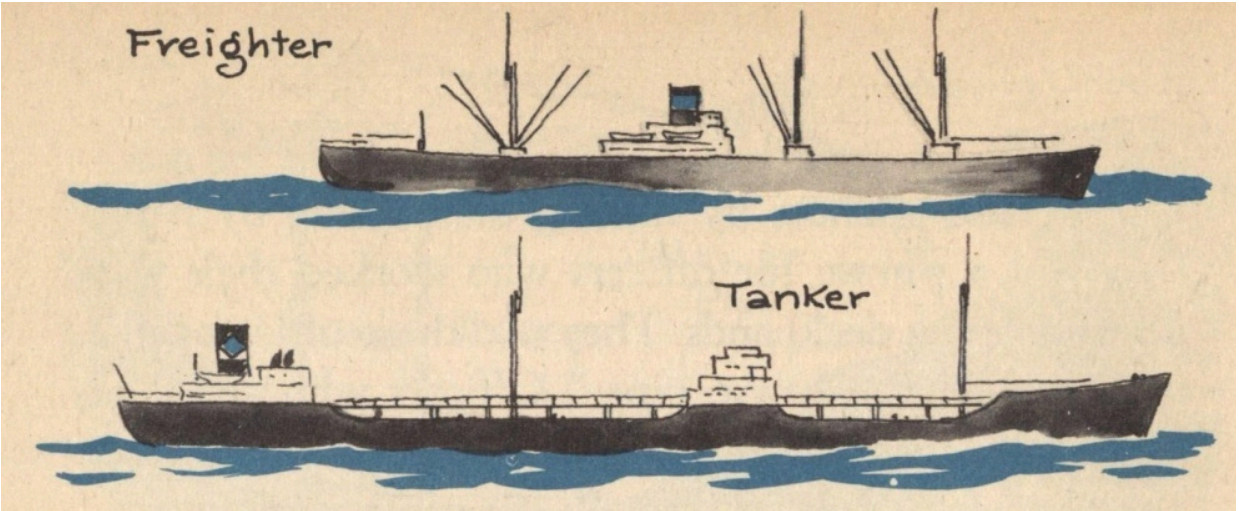
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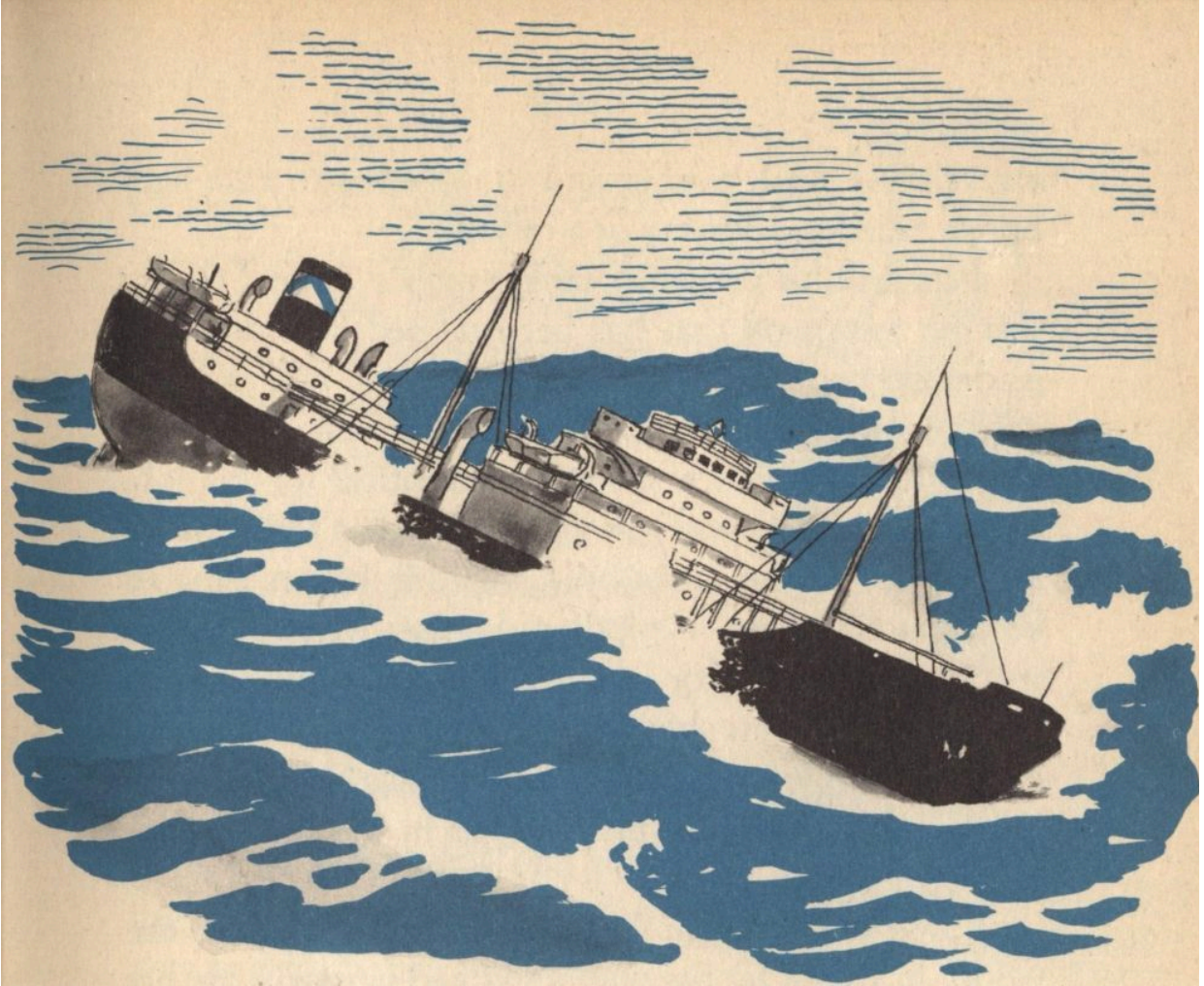
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Outrigger
of the
Sulu Sea



Junk



Chinese Tub-boat



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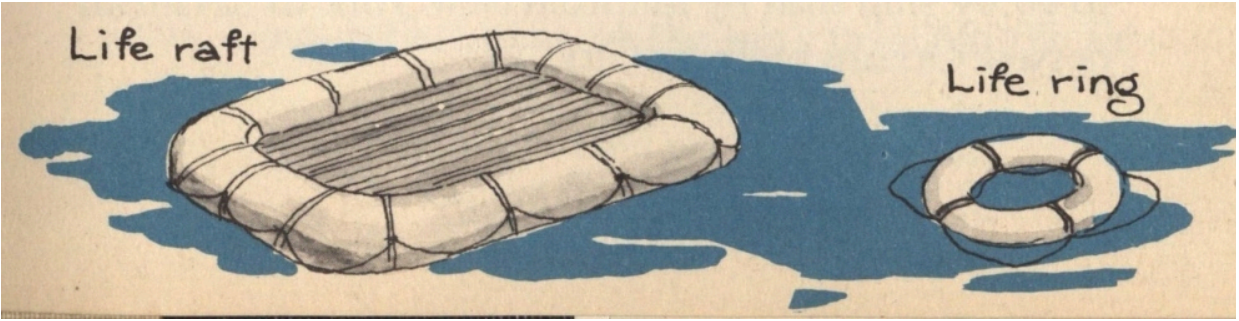


A Quffa on the Tigris

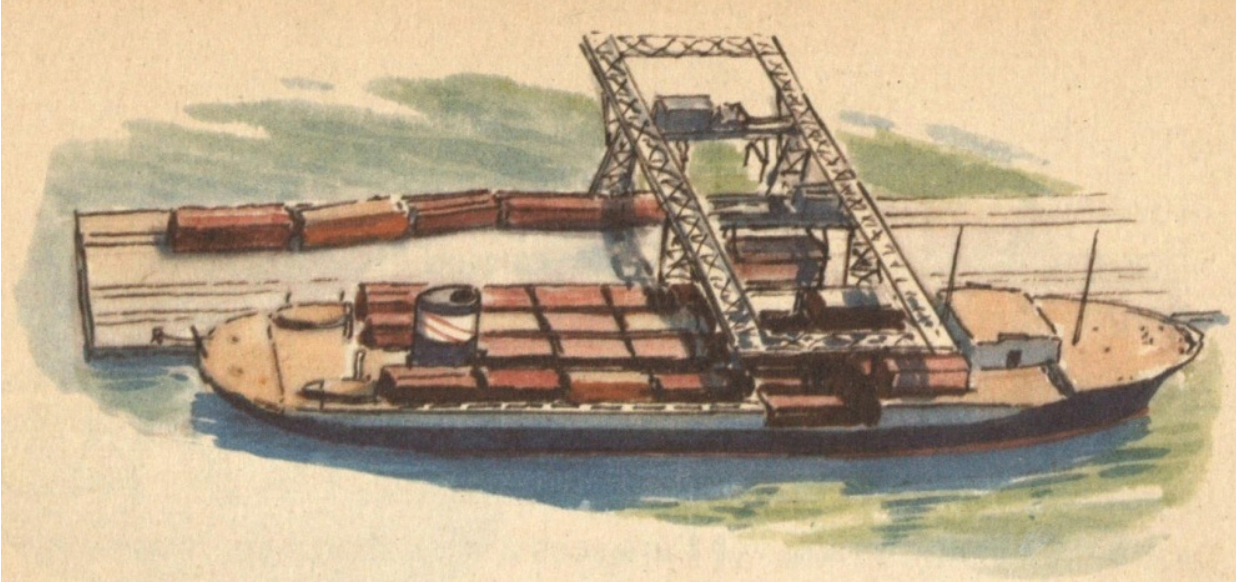


Raft of Timber and Inflated Skins Discharging Grain at Bagdad. Small quffas Serve as Lighters.

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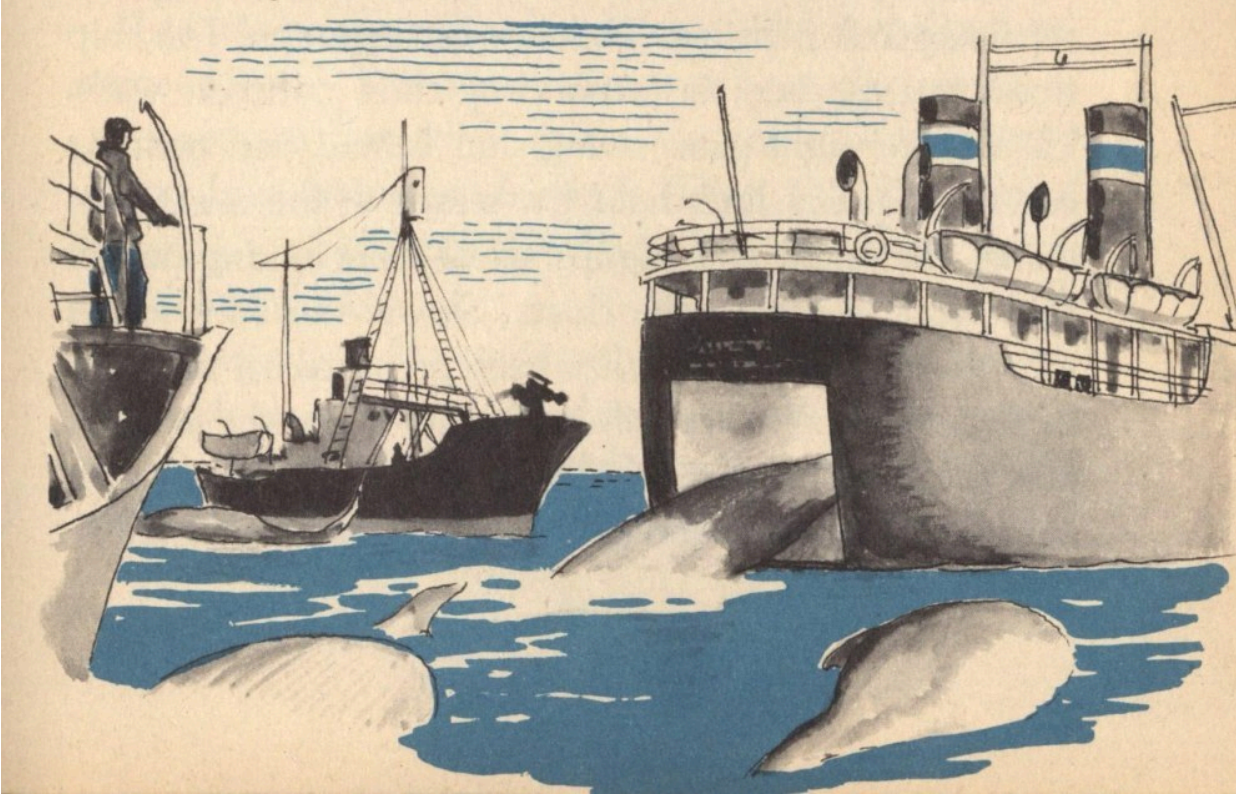
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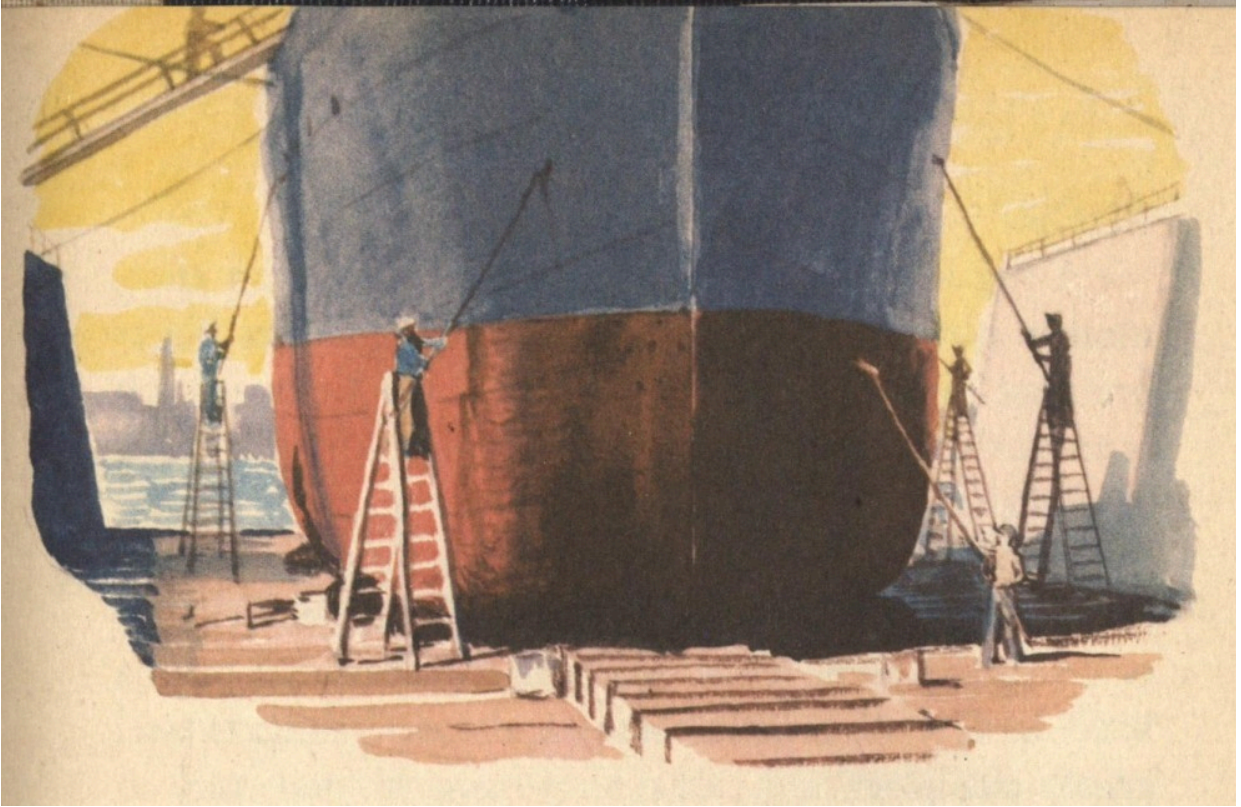
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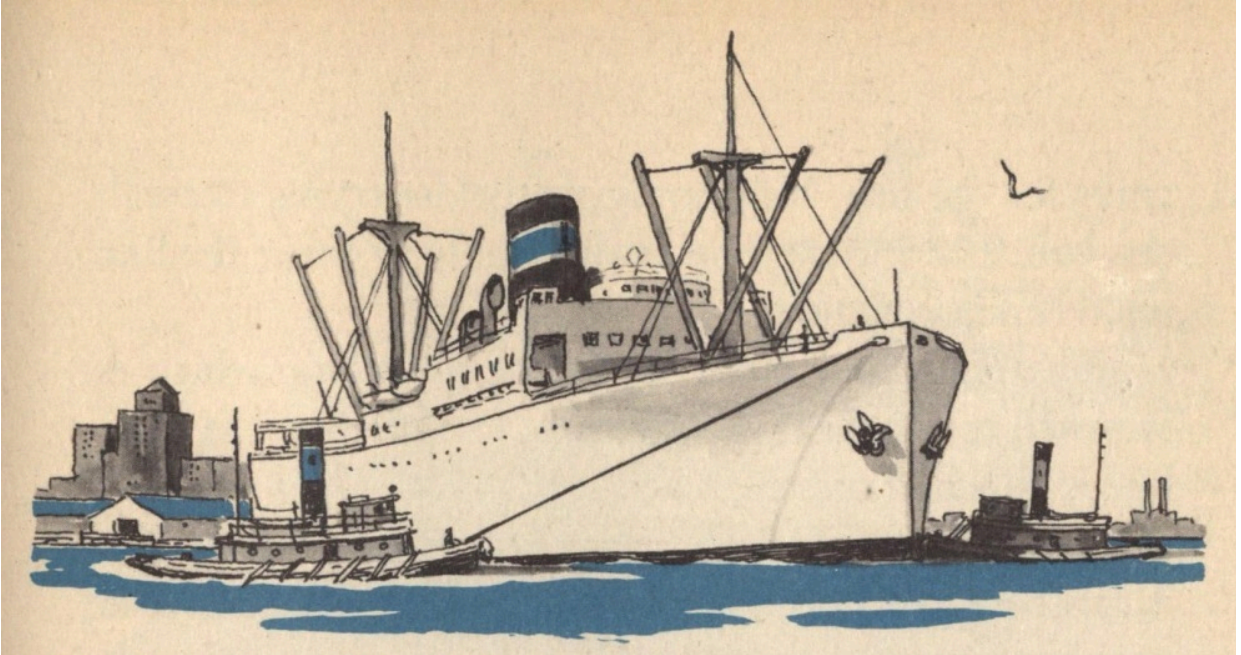
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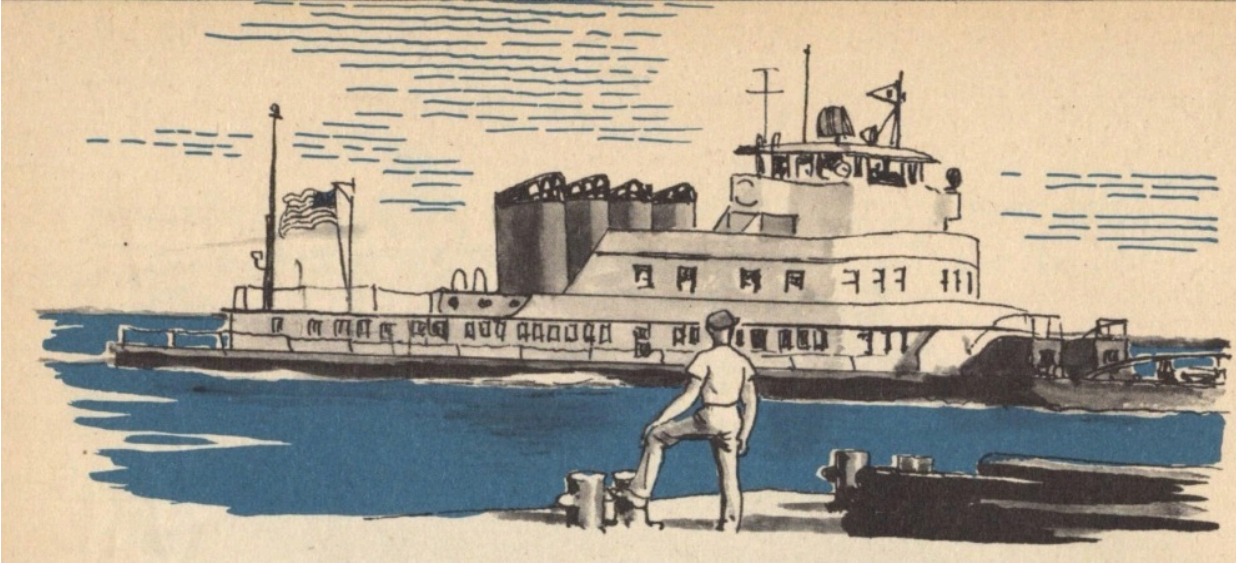
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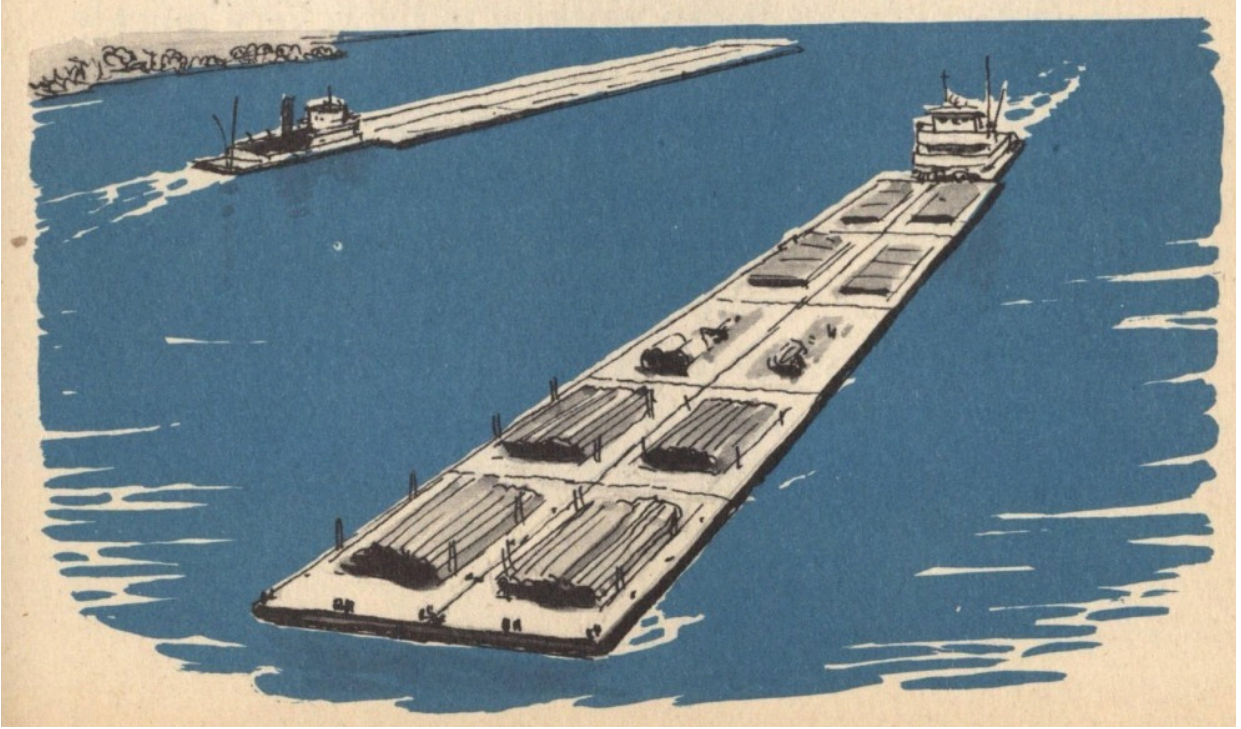
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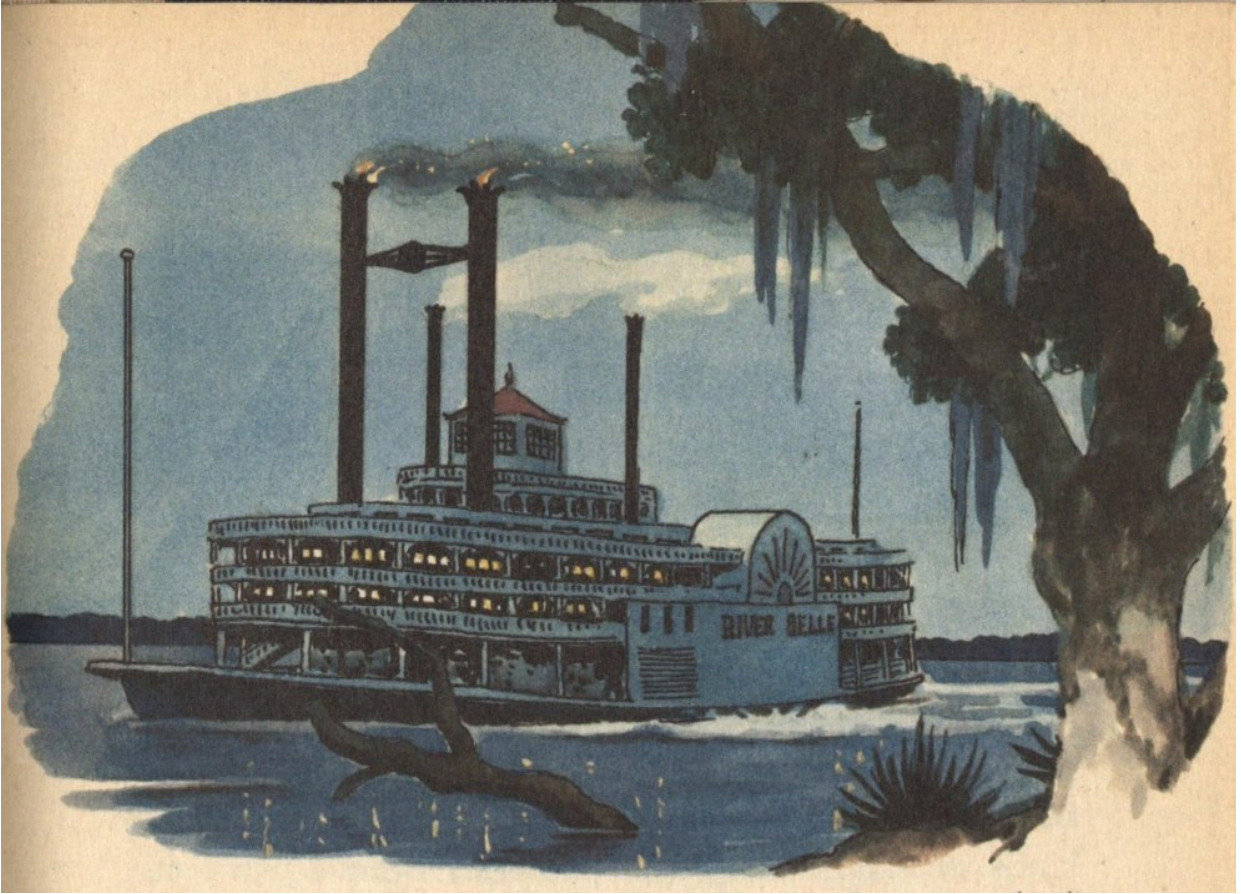
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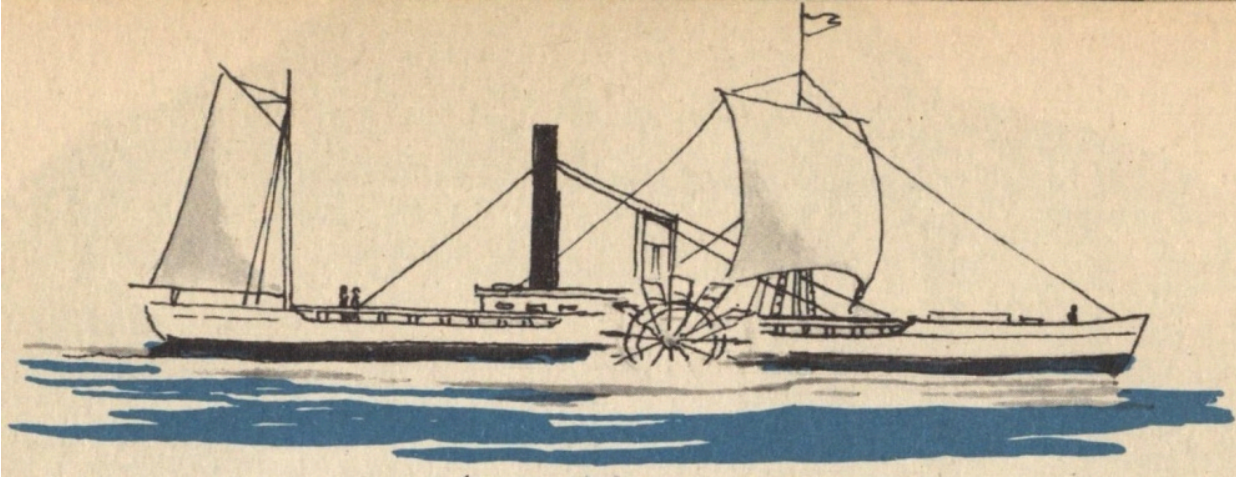
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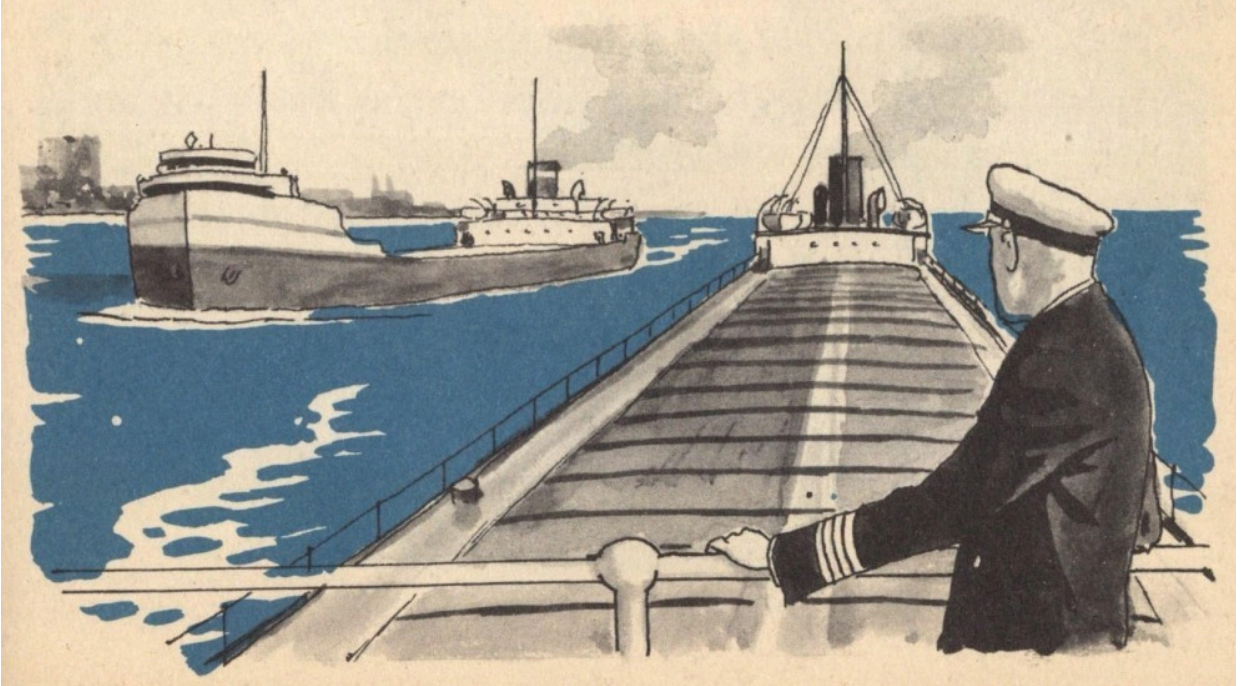
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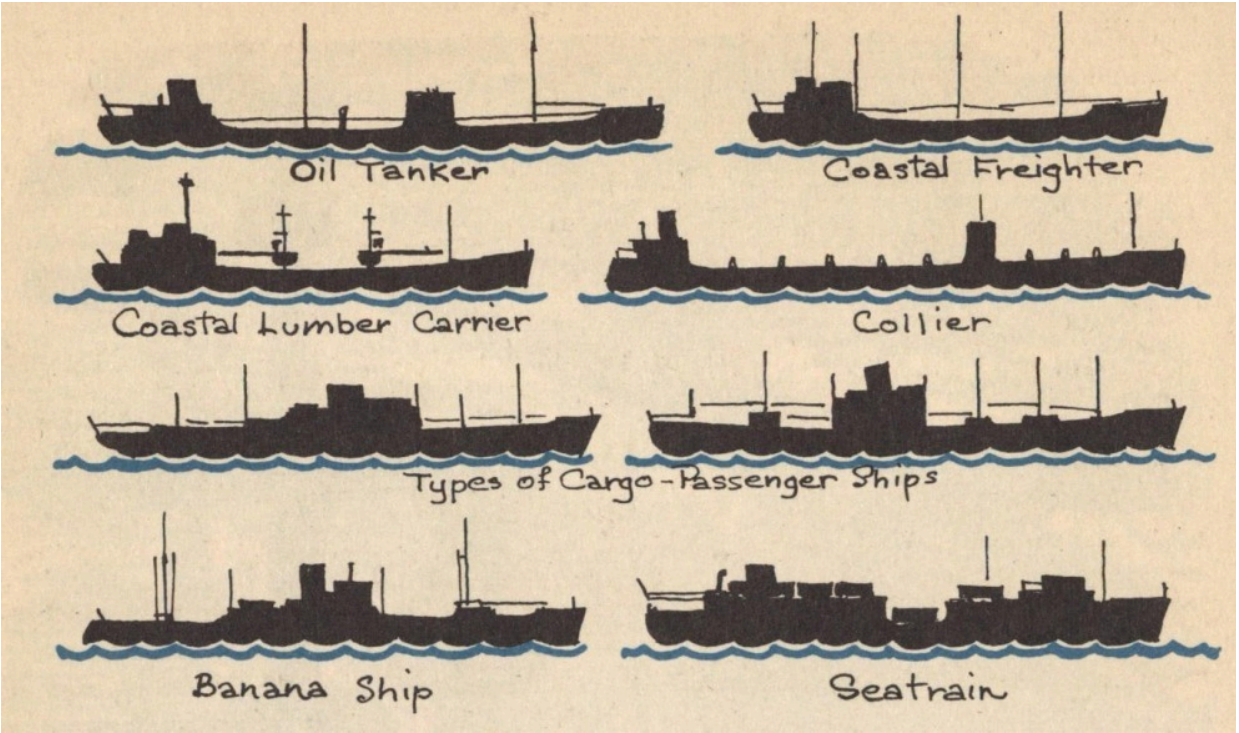
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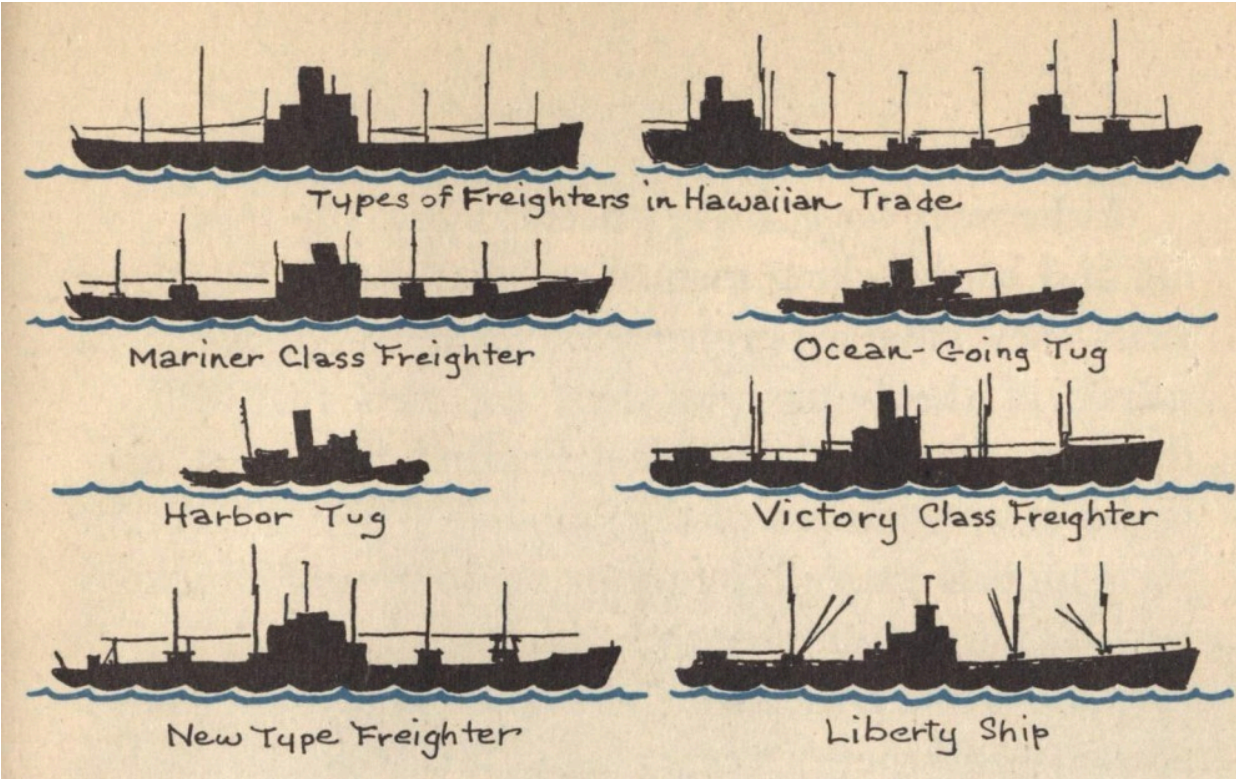
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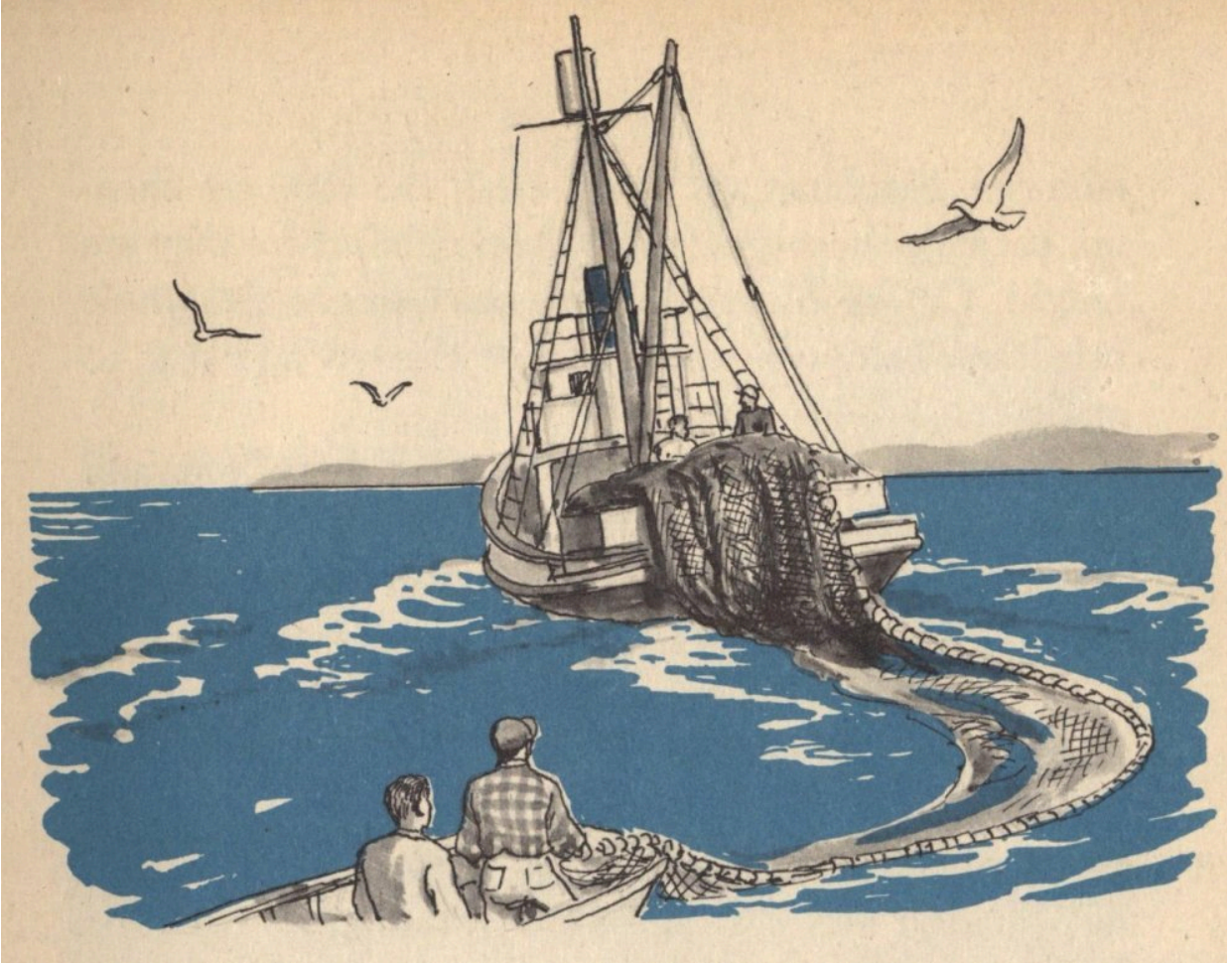
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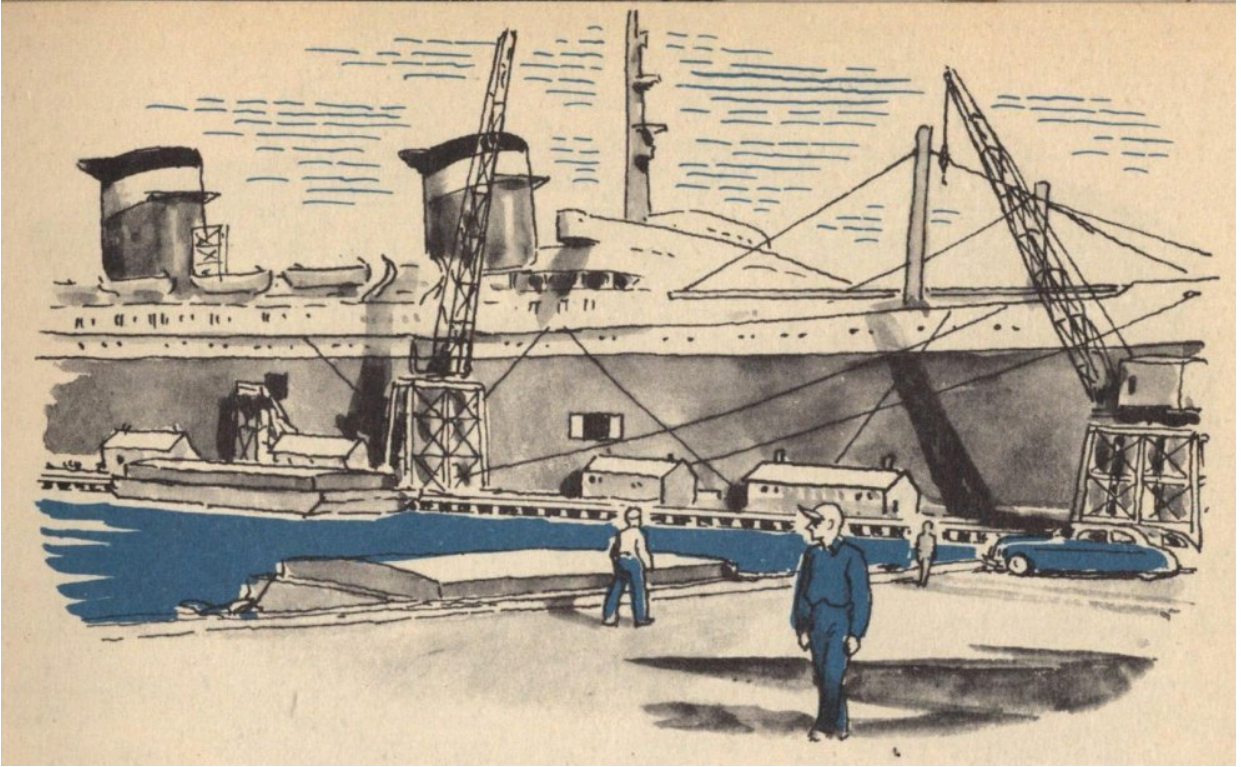
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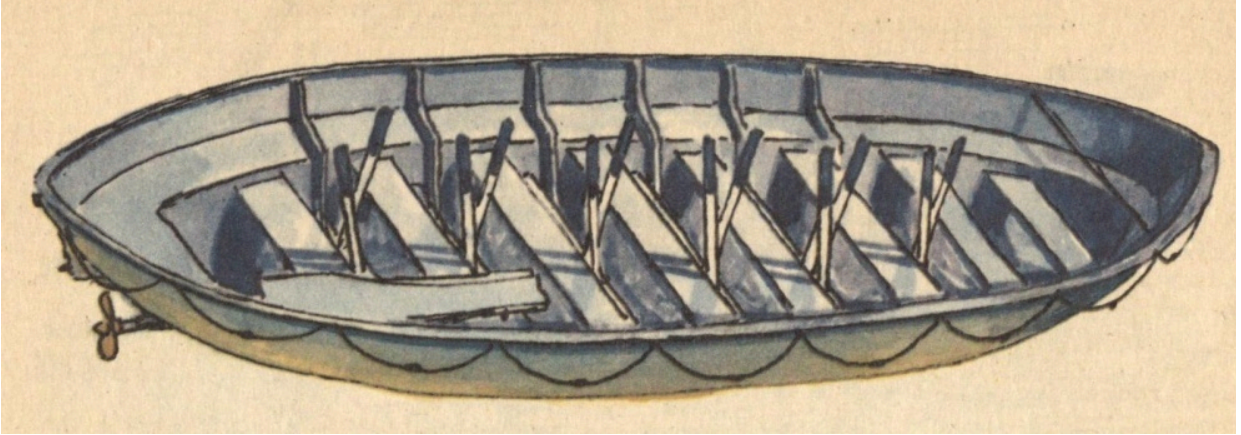
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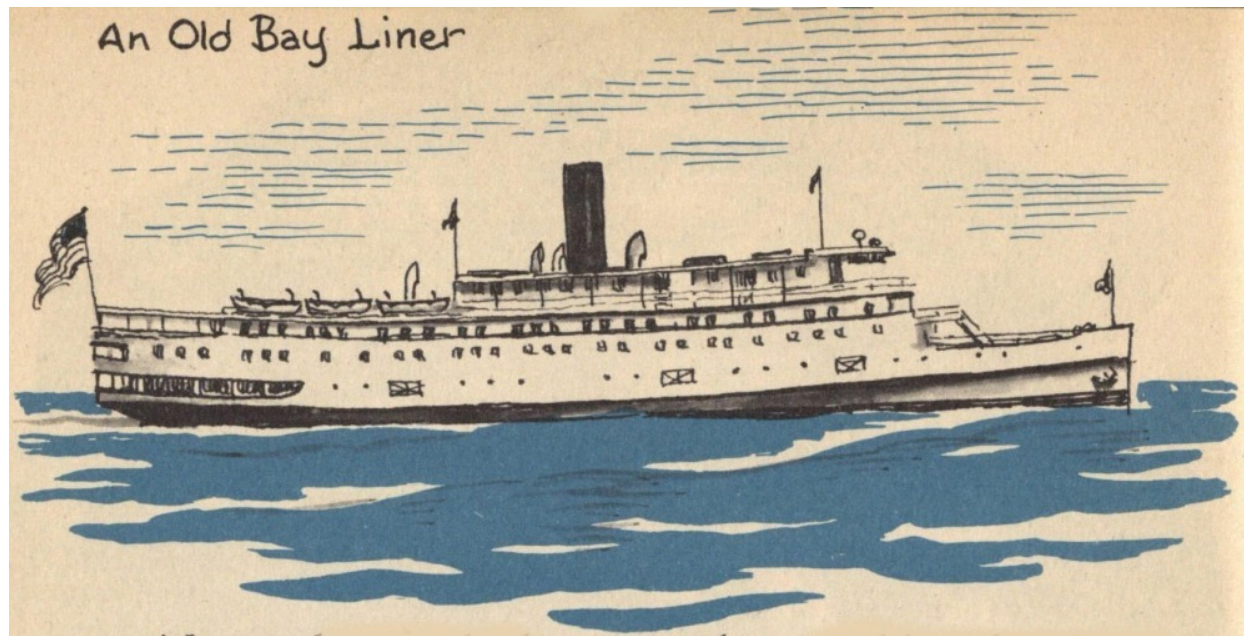
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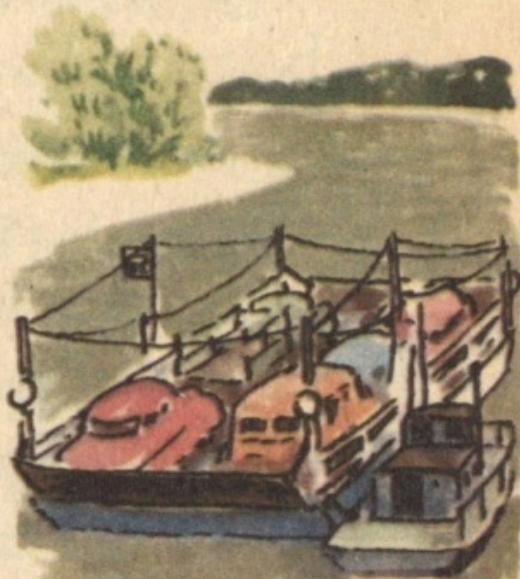
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Staten Island Ferry

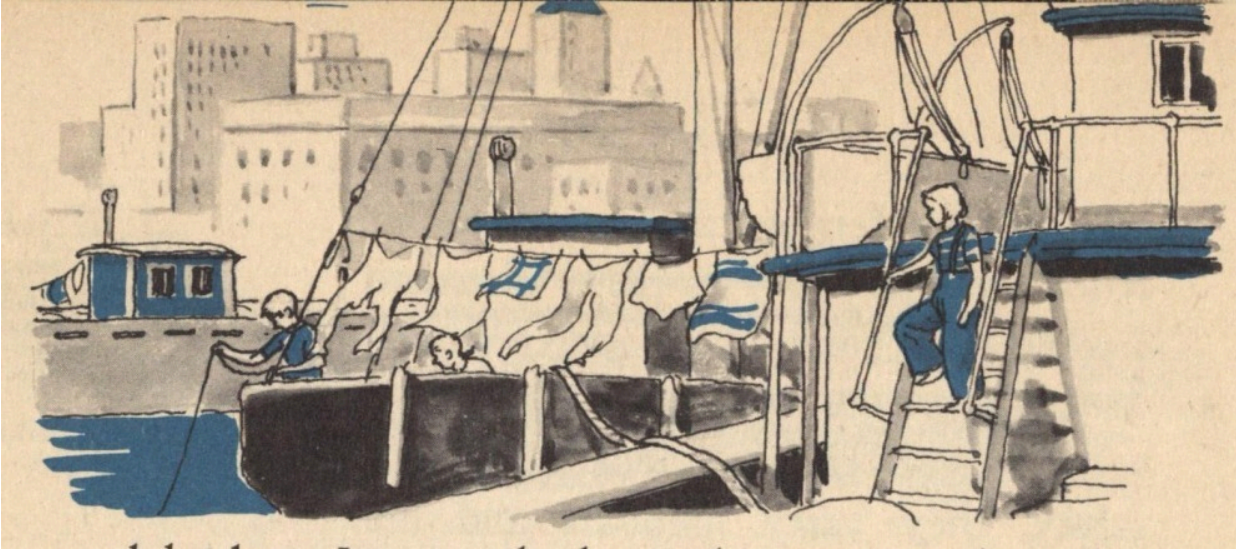


Arkansas
River Ferry

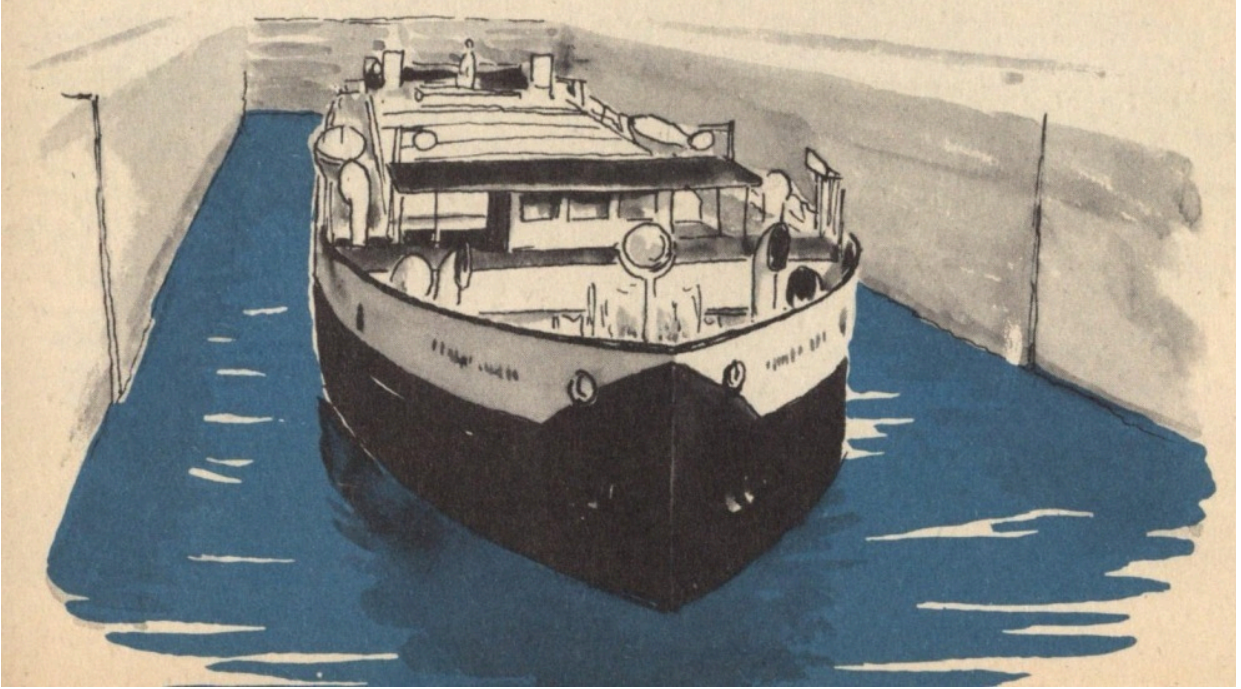
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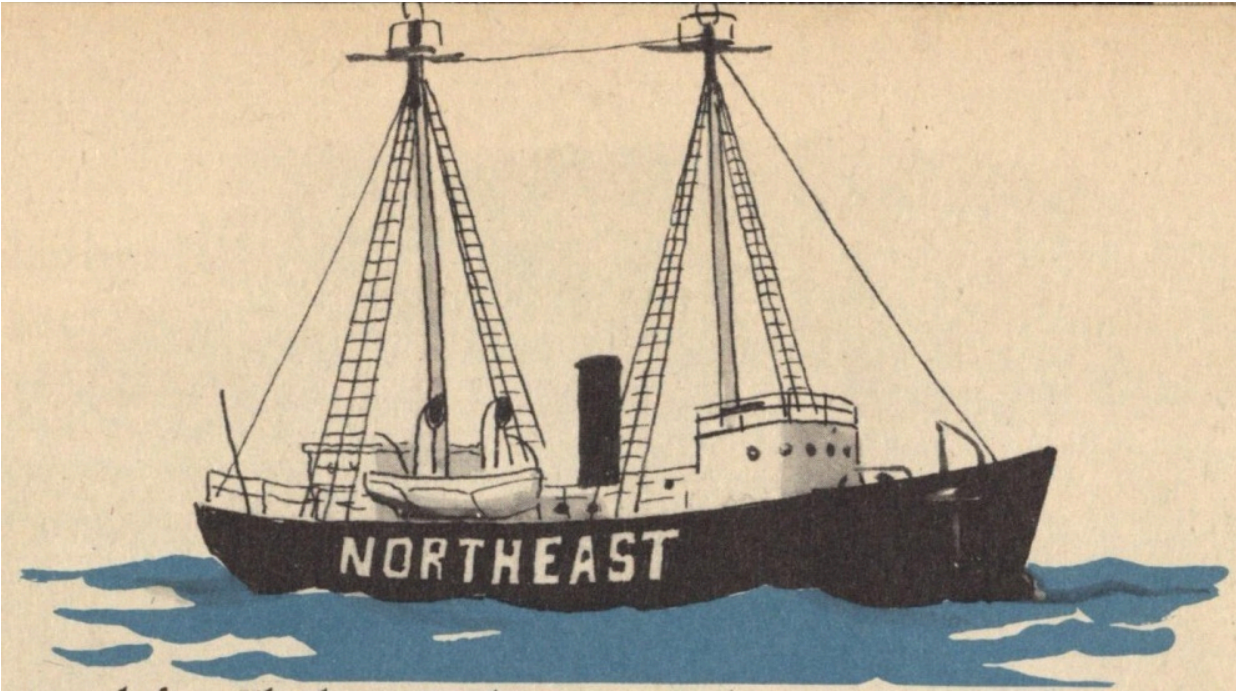
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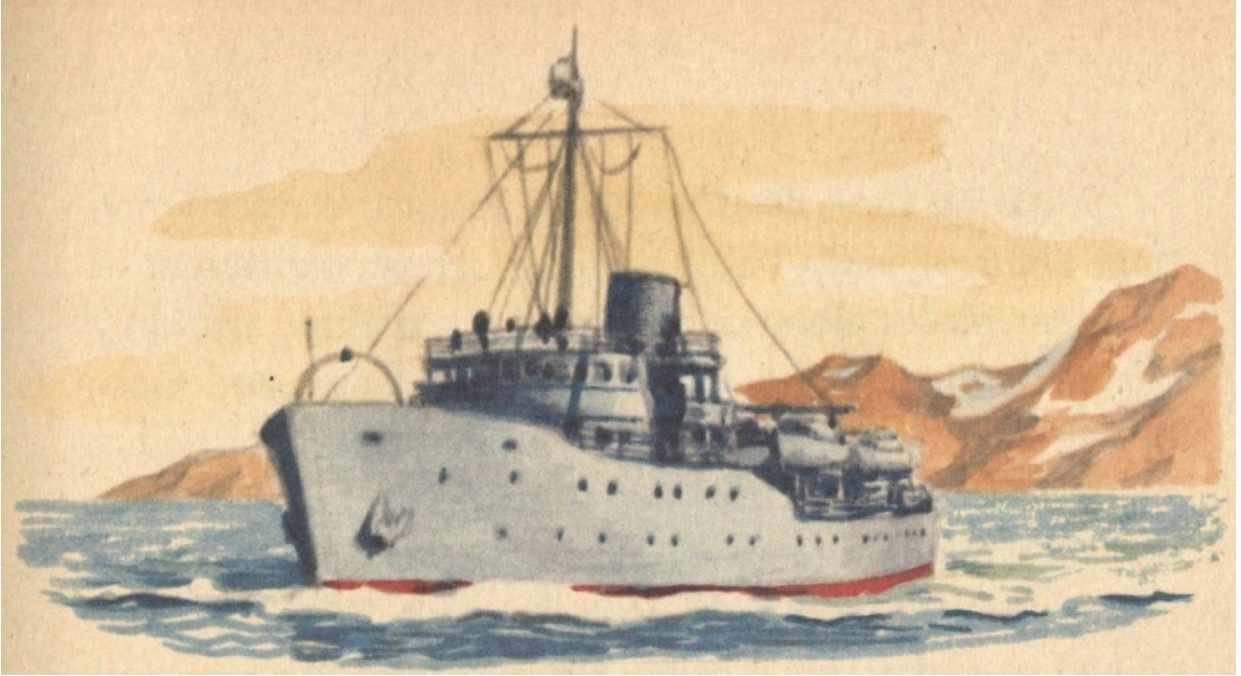
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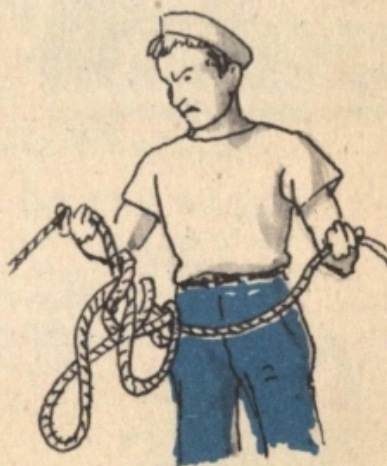
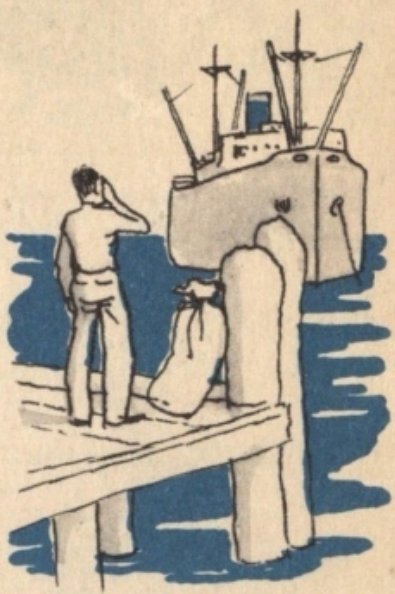
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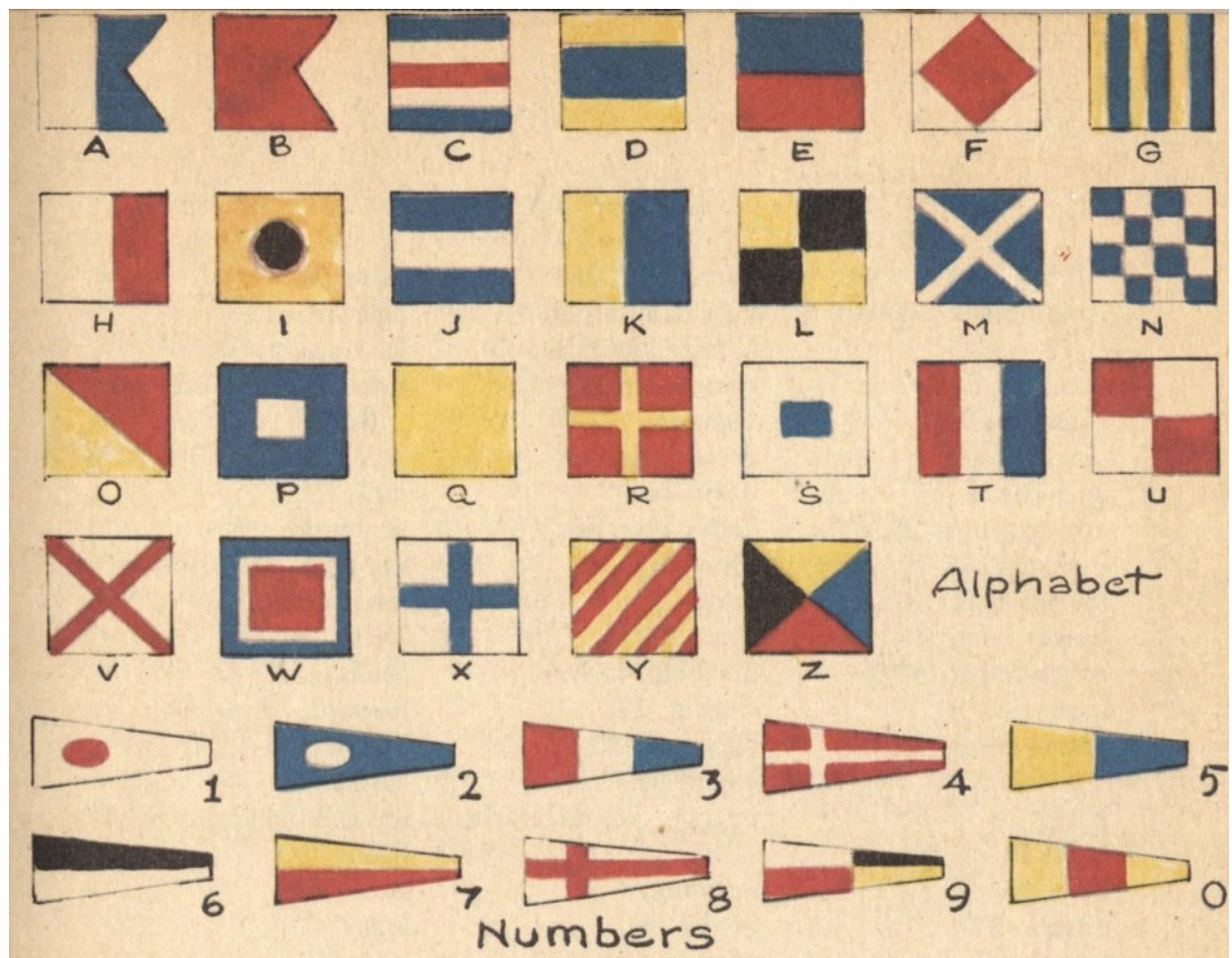
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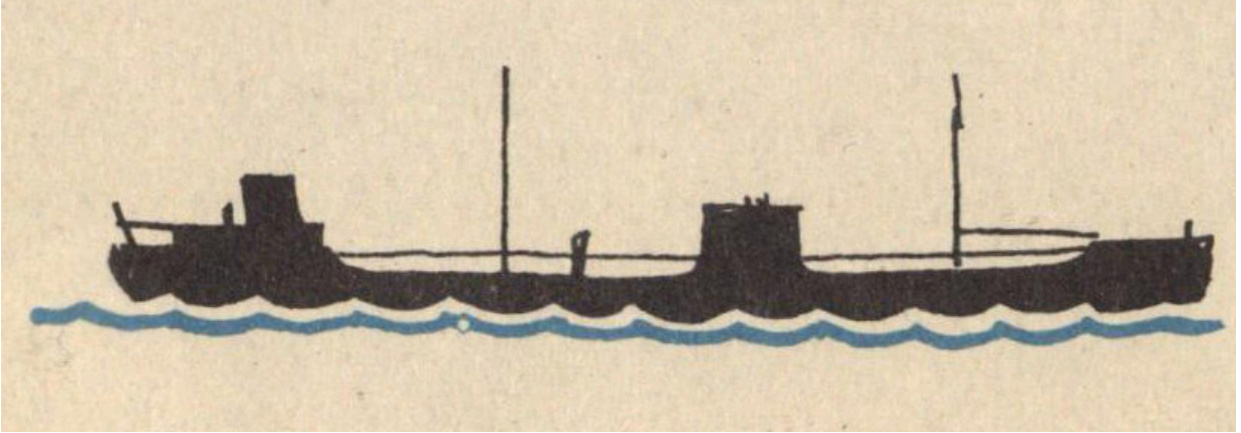
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